How to (and How Not to) Increase Free-Shipping Thresholds: Evidence from Online Retail

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Abstract. **Problem definition:** Contingent free shipping (CFS) policies are widely used by e-commerce platforms to encourage larger cart values, yet platforms increasingly struggle with how to adjust these thresholds in response to rising fulfillment costs. This paper examines the impact of a history-dependent threshold increase on consumer behavior and derives conditions under which platforms can increase the thresholds. **Methodology:** Using detailed transaction and clickstream data from a North American online grocery platform, we analyze a threshold increase from $80 to $100 by employing a difference-in-differences design. **Results:**We find that, among continuing purchasers, raising the CFS threshold results in a 19.4% decline in consumer spending, driven primarily by a reduction in average cart value and, to a lesser extent, a decline in purchase frequency. More importantly, the decline is larger among consumers who were more likely to have adjusted their carts to clear the original threshold and among those who purchased more frequently or more recently under the original threshold. We propose reference-dependent perceived attainability as a potential mechanism: the original threshold serves as a reference point, making the new threshold feel less attainable and weakening CFS incentives. We provide further evidence for this mechanism using a randomized online controlled experiment. **Managerial implications:** To assess the implications for CFS threshold revision, we develop a stylized analytical model that captures both cost- and demand-side factors associated with increasing the CFS threshold. We find that it is optimal to increase the CFS threshold only when the average cart value is sufficiently close to the original threshold. This result aligns closely with our empirical findings and remains robust to extensions that incorporate consumer heterogeneity.

Key words: e-commerce, contingent free shipping, consumer purchase behavior, difference-in-differences

1. Introduction

E-commerce platforms have experienced explosive growth over the past decade. As of 2025, 2.77 billion consumers worldwide shopped online, generating more than $6.3 trillion in sales, accounting for 20.1% of global retail purchases (Seller Commerce 2025a). This share is projected to reach 25% by 2030 (Statista 2025). Despite these impressive growth figures, online retailers continue to face a persistent challenge: converting website visitors into paying consumers. Many potential buyers abandon their carts before completing a purchase. A leading reason for checkout abandonment is the presence of additional costs, most notably shipping surcharges (Baymard Institute 2025).

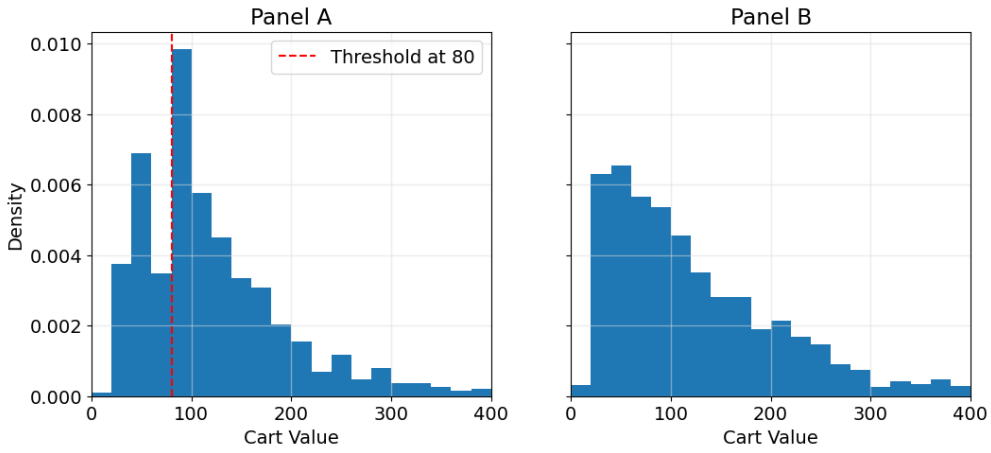
To address this challenge, online retailers use different shipping policies to balance delivery costs and demand. While some charge a shipping fee (either a flat fee regardless of order size or distance-based/weight-based fees that vary with logistics costs), others offer unconditional free shipping for all consumers or only for paid members. A third shipping policy that is becoming popular is contingent free shipping (CFS), where the retailer waives the shipping fee only if a consumer's order exceeds a predefined spending threshold.

The rationale behind CFS policies is twofold. From a cost perspective, aggregating larger orders allows online platforms to reduce per-unit delivery costs through economies of scale. From a demand perspective, by offering free delivery contingent on meeting a spending threshold, CFS policies incentivize consumers to add items to their carts or switch to higher-priced products. The effectiveness of CFS in shaping consumer purchase behavior has been documented in industry practice (Brandt 2018) and examined in academic research (Cachon et al. 2018, Chen and Ngwe 2018, Hemmati et al. 2021, Kabra et al. 2024, Lewis et al. 2006). Given these benefits, it is no surprise that approximately 65% of online retailers today employ some form of contingent free shipping to reduce shipping frictions and attract more consumers (Seller Commerce 2025b).

However, to manage rising fulfillment costs, many of these platforms periodically raise the free-shipping threshold (Sancia 2025), with the average CFS threshold rising from approximately $52 in 2019 to $103 in 2025 (Nigrelli 2025). From a cost perspective, such adjustments are well justified: increasing the threshold reduces the share of orders in which the platform bears shipping costs. Their demand-side consequences, however, are less straightforward. Recent policy reversals illustrate the challenges of revising an established CFS policy. Amazon Fresh increased its free-shipping threshold from $35 to $150 in February 2023, only to reduce it to $100 ten months later (Moran 2023). Similarly, Freshippo, an online grocery platform operated by Alibaba Group, raised its threshold from 49 RMB to 99 RMB in February 2024, but reverted to the original level within two months (Chen 2024). These reversals highlight the practical importance of understanding how consumers' experience and purchasing adjustments under an established CFS threshold shape their response when that threshold is subsequently raised.

Motivated by this gap, the central objective of this paper is to study the demand-side effects of increasing CFS thresholds. We address three interrelated research questions: *(1) What is the impact of a history-dependent threshold increase on consumers? (2) What behavioral mechanisms underlie the history-dependent response to a threshold increase? (3) What does the history-dependent response imply for revising an established CFS policy?*

Figure 1 Cart Value Distributions with CFS (Panel A) and without CFS (Panel B) policies.



Notes: All completed carts are observed between January 2023 and October 2023. Panel A shows cart values for metropolitan regions under a CFS policy with an $80 threshold, while Panel B shows cart values for non-metropolitan regions without a CFS policy.

To answer these questions, we analyze a unique panel dataset from our industrial partner, which is a North American e-commerce platform that connects more than 1,000 farmers and food processors with end consumers. Like many online grocery retailers, this platform faced high delivery costs. To exploit economies of scale in fulfillment, the platform introduced a CFS policy for consumers living in metropolitan areas, offering them free delivery for orders that exceed $80. In contrast, consumers residing elsewhere are subject to standard paid shipping.

Using cart-level transaction data from the collaborating platform, we confirm that consumers adjust their carts with the original threshold in our empirical setting. Figure 1 presents the distribution of completed cart values under these two regimes. Panel A shows a pronounced mass of orders just above the $80 free-shipping threshold in CFS regions, accompanied by relatively fewer orders immediately below $80. In contrast, Panel B shows no such concentration when free shipping is unavailable. These patterns align with prior studies, indicating that the existing CFS threshold effectively encourages consumers to increase their cart values.

Nevertheless, the platform increased the CFS threshold from $80 to $100 in November 2023 for consumers in metropolitan regions, while leaving shipping policies unchanged elsewhere. This policy change creates a natural difference-in-differences (DiD) setting spanning nearly two years. Namely, between January 2023 and October 2023, consumers in metropolitan regions faced an $80 threshold; from November 2023 onward, the threshold increased to $100. In addition to transaction-level purchase data, we observe detailed clickstream data, which enabled us to examine the behavioral mechanisms underlying consumers' responses.

To answer the first research question, we document several novel findings using the DiD identification strategy. Following the threshold increase, consumers who remain active on the platform reduce spending by 19.4%, driven primarily by a 19.0% reduction in average cart value and, to a lesser extent, a 3.1% decrease in purchase frequency. The composition of consumers' baskets also shifted: consumers purchased 7.3% fewer distinct products, switched to lower-priced items, with unit prices declining by 11.5%, and reduced the share of spending allocated to popular products by 3.3 percentage points.

To answer the second research question, we provide evidence consistent with a mechanism of reference-dependent perceived attainability. Drawing on reference dependence and goal-gradient theory, we propose that repeated shopping under the original CFS policy may help establish the original threshold as a benchmark for qualifying for free shipping. More recent shopping experiences may make this benchmark easier to recall. When the threshold increases, comparison with this benchmark can make the new requirement seem farther away and less attainable, weakening consumers' willingness to engage in CFS-induced cart adjustment. Consistent with this mechanism, the decline is larger among consumers who adjusted their carts more often to clear the original threshold before the increase. The decline is also larger among those who purchased more frequently or more recently. A randomized online controlled experiment provides further complementary evidence: with current purchase requirements held constant, participants presented with a lower prior threshold report lower perceived attainability and are less likely to choose to adjust their carts.

To close the loop and to answer the third research question, we develop a stylized analytical model to study the implications of our empirical findings for optimal free-shipping threshold design. The model integrates both cost-side and demand-side forces that govern the profitability of adjusting the CFS threshold. We first characterize the platform's optimal threshold policy and show that it exhibits a *bang–bang* structure: it is optimal to increase the CFS threshold only when the associated observed average cart value lies within a feasible range, and optimal not to increase it otherwise. This result aligns closely with our empirical evidence, which indicates that consumers' CFS-induced behavior only operates within a certain range. Once the threshold exceeds this range, the resulting loss in demand outweighs the cost savings from reduced shipping subsidies. We also extend the stylized model to incorporate consumer heterogeneity with respect to the distribution of cart values. We show that the *bang–bang* nature of the optimal threshold policy remains, which reinforces the generalizability of our analytical insights.

Our findings yield several managerial implications. First, managers often rely on observed average cart values under the existing CFS policy to justify higher thresholds (e.g., Shopify 2025, USPS Delivery 2025). We show that this metric can be misleading, as it reflects endogenous cart adjustment behavior rather than the underlying consumer demand. A higher observed average cart value may, counterintuitively, indicate that the threshold should not be increased. Second, we propose a staggered one-time rollout strategy: The platform selects a higher threshold and a cutoff for the time elapsed since a consumer's last purchase. Each consumer continues to face the original threshold until the interval exceeds the cutoff. Our simulations suggest that this strategy may mitigate the adverse impact of threshold increases.

The remainder of the paper is organized as follows. § 2 reviews the related literature. § 3 describes the research context and data. § 4 presents the treatment effects of the threshold increase. § 5 investigates the underlying mechanisms. § 6 discusses managerial implications, and § 7 concludes.

2. Literature Review

Our study relates to two streams of research: (1) studies examining the effects of shipping policies and services on consumer behavior and (2) research on optimal CFS policy design. We contribute to the first stream by examining how consumers' experience and purchasing adjustments under an established CFS threshold shape their response when that threshold is raised. We contribute to the second by examining how these responses inform a retailer's decision to retain or increase an existing threshold.

2.1. Shipping Policies and Purchasing Behavior

Shipping policies and services influence several aspects of consumer purchasing behavior, including purchase incidence, spending, basket size, purchase frequency, and product returns. Prior research has examined these effects in relation to both price-related attributes, such as shipping fees and free-shipping thresholds (Chen and Ngwe 2018, Lewis 2006, Shehu et al. 2020), and non-price attributes, such as delivery speed, reliability, and convenience (Amorim et al. 2024, Cui et al. 2020, 2024, Fisher et al. 2019, Hu et al. 2024, Lu et al. 2025). Of particular interest to our study is research on shipping fee policies, especially CFS, which links shipping charges to order values and influences consumers' purchasing decisions.

Early empirical studies establish that shipping charges affect both purchase incidence and order size. Lewis (2006) examines customer acquisition, repeat ordering, and purchase quantities, while

Lewis et al. (2006) estimate heterogeneous consumer responses to nonlinear shipping-fee schedules. Their analyses show that waiving shipping fees for larger orders can encourage larger baskets, whereas unconditional free shipping can increase purchase incidence while reducing average order size. These findings establish that the structure of shipping charges matters for purchasing. Consumers' evaluations of shipping fees also depend on whether they view these changes as a way for retailers to recover delivery costs or generate additional profit (Koukova et al. 2012).

Subsequent research examines how shipping incentives interact with the shopping environment. Chen and Ngwe (2018) study multicategory basket formation, showing that CFS can generate larger baskets than unconditional free shipping and shift demand toward more popular products. Their analysis also links search costs to consumers' difficulty in meeting the threshold without overshooting. Jin et al. (2023) document differences in order padding between PC and mobile channels. Shipping policies also affect product returns and channel choice: Shehu et al. (2020) show that unconditional free-shipping promotions encourage purchases of more risky products and increase returns, while Hemmati et al. (2021) investigate CFS-induced padded orders with elevated return propensities. Kanuri et al. (2025) find evidence of order aggregation and substitution toward offline purchasing after changing from tiered shipping charges to a flat per-order fee.

Among studies examining shipping fee policies, Cachon et al. (2018) and Kabra et al. (2024) are particularly relevant because both use data from online retailers that revised their CFS terms. Cachon et al. (2018) develop a data-driven model to assess the profitability of alternative thresholds, accounting for order padding and subsequent product returns at a U.S. apparel retailer. In their basket-formation model, padding depends on the gap between base demand and the threshold currently in effect. They estimate the model using transactions under the original high threshold and assess its predictions after the retailer lowers the threshold to one-third of its original level. They also separately examine aggregate daily sales, finding no statistically significant sales effect, while observed average completed-order value declines after the threshold reduction. Their transaction data do not contain customer-level information, so baskets cannot be linked to examine whether individual consumers' adjustments under the original threshold shape their response to its revision.

Kabra et al. (2024) examine how the ease of returning products moderates CFS-induced order-swelling and return-swelling. Their online fashion retailer first raises the threshold while leaving the shipping fee unchanged and subsequently raises the fee. Both changes apply to all markets simultaneously. Their difference-in-differences design compares responses in markets with and without convenient return pickup. As a result, their research question concerns how the return environment

conditions CFS performance, rather than how prior-threshold experience shapes consumers' evaluation of a revised requirement. They find that return convenience affects both order-swelling and return-swelling, with the latter playing a central role in differences in policy profitability across markets.

In contrast, our study examines how prior-threshold experience shapes consumers' response to a threshold increase in an online grocery setting without product returns. The focal platform raises an established threshold in all CFS-enabled regions while leaving shipping policies unchanged in other regions. The shipping-fee schedule below the threshold does not change. This variation supports a consumer-level difference-in-differences analysis that follows purchasing before and after the increase and compares consumers facing the revision with those whose policy remains unchanged. Among continuing purchasers, we estimate declines in monthly spending, average cart value, and the monthly number of completed transactions. The linked product and clickstream records further allow us to relate subsequent purchasing responses to basket composition and consumers' prior threshold-related shopping patterns.

We examine reference-dependent perceived attainability as a potential mechanism underlying this history dependence. Building on reference dependence and goal-gradient research (Kahneman and Tversky 1979, Kivetz et al. 2006, Tereyağoğlu et al. 2018), we consider whether the original threshold serves as a benchmark against which consumers evaluate the attainability of the revised requirement. Lewis (2006) investigates reference effects associated with previous shipping charges, but reports inconclusive evidence. In our setting, the proposed reference point is the earlier free-shipping threshold. Similarly, although Kanuri et al. (2025) connect policy responses to prior purchasing behavior, their mechanism concerns order aggregation under a new flat-fee incentive, rather than the role of an earlier CFS threshold in evaluating a new one.

2.2. Optimal CFS Policy Design

This stream of research treats CFS design as part of a broader pricing and operations problem. Leng and Becerril-Arreola (2010) jointly determine product pricing and free-shipping thresholds in monopoly and duopoly settings. Becerril-Arreola et al. (2013) incorporate inventory decisions into a simulation-based analysis of promotional pricing and thresholds. Li et al. (2023) jointly optimize product margins, below-threshold shipping fees, and free-shipping thresholds, characterizing how consumer heterogeneity affects the optimal policy. Related work by Shao (2017) compares free and calculated shipping in competitive supply chains, highlighting the role of shipping costs and channel structure rather than optimizing a CFS threshold. These studies show that effective shipping-policy

design requires balancing purchasing incentives, shipping-cost recovery, and other pricing and operational decisions.

The consumer-response assumptions underlying these decisions are particularly relevant to our study. Li et al. (2023) allow consumers to top up, purchase below the threshold and pay for shipping, qualify without adjusting their basket, or make no purchase. A higher threshold can therefore induce some consumers to buy more while deterring others from topping up. Likewise, the basket-formation model of Cachon et al. (2018) allows padding to become less likely as the gap between base demand and the threshold increases, which can generate a nonmonotonic relationship between the threshold and average basket value. Thus, existing models already accommodate the possibility that raising a threshold reduces purchasing. Although some models also incorporate repeat purchasing, their threshold-dependent responses are specified through the policy terms currently offered, rather than consumers' experience under the threshold being replaced.

Our contribution to this literature is an empirically informed *conditional threshold revision decision*. Taking the existing threshold and the cart value observed under that policy as our starting point, we use the estimated purchasing response to evaluate whether to retain or increase the threshold. Because the observed baseline may already reflect CFS-induced purchases, it should not be interpreted as policy-free demand. Our analysis shows that the attractiveness of a threshold increase depends on observed cart value relative to the existing threshold: an average cart value above that threshold does not, by itself, justify an increase. Our findings inform whether and how retailers should revise an established CFS threshold.

3. Research Context and Data

3.1. Research Context

To examine consumer responses to the CFS threshold increase, we leverage a policy change implemented by a North American platform. Founded in 2019, the platform operates as an online grocery retailer. It does not own or operate brick-and-mortar stores, nor does it serve as a delivery intermediary that fulfills orders from conventional supermarket chains. Instead, it procures products directly from more than 1,000 independent suppliers, including local farmers and food processors. Among them, only a small share operate their own online sales channels or distribute their products through conventional supermarket chains or franchises. The platform purchases products directly from these suppliers, temporarily holds the inventory in a single warehouse, and uses that facility to assemble and dispatch customer orders. The platform serves customers across 450 locations and offers approximately 11,000 products spanning nearly 400 subcategories.

We provide descriptive statistics to better understand the consumer base. From January to October 2023, 1,431 consumers placed 3,683 paid orders. The median purchaser placed one order, compared with a mean of 2.57 orders over ten months. Although 71.8% of these purchasers placed only one order, repeat purchasers generated 72.1% of all paid orders. Products classified as groceries account for 81.5% of the total revenue, with substantial purchasing in prepared foods, meat and poultry, produce, pantry products, dairy and eggs, bakery products, and seafood.

The platform applies different shipping policies based on consumers' location. Consumers residing in metropolitan regions are eligible for the CFS policy, whereas others are subject to standard paid shipping. At checkout, consumers eligible for the CFS policy are shown not only the shipping surcharge based on their current cart value but also the additional amount they need to spend to qualify for free shipping. In contrast, consumers outside the CFS-entitled regions are shown only the applicable shipping surcharge. Among completed orders with positive shipping payments, fees range from \$11 to \$13 for CFS-enabled consumers, with a mean of \$11.73, and from \$11 to \$24 for other consumers, with a mean of \$15.68. Shipping fees vary across predefined shipping zones, with each consumer's applicable fee determined by their delivery address.

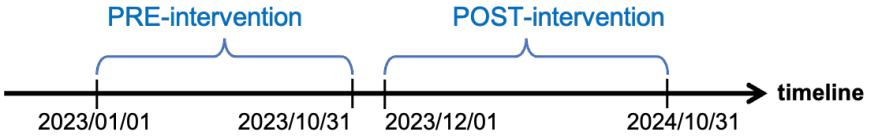
The free shipping threshold under the CFS policy was initially set at \$80, and raised to \$100 in November 2023 for all consumers in the CFS-entitled regions. Moreover, the adjustment was implemented without prior announcement, promotional campaigns, or any concurrent policy change. This policy shift provides a quasi-experiment setting, allowing us to isolate and identify the behavioral impact of the threshold increase without confounding influences from other marketing interventions.

3.2. Data

We obtained consumer-level *transaction data* and *clickstream data* from the collaborating platform, covering the period from January 1, 2023, to October 31, 2024. As shown in Figure 2, we divide the sample into three distinct phases: (1) the pre-intervention period (January 2023 – October 2023), (2) the intervention period (November 2023), during which the platform implemented and rolled out the new threshold, and (3) the post-intervention period (December 2023 - October 2024). We restrict our primary analysis to the pre- and post-intervention periods and focus on consumers who placed at least one order in both periods. This restriction, commonly employed in consumer-level analyses (e.g., Guo and Liu 2023, Lu et al. 2025), allows for meaningful within-consumer comparisons of pre- and post-intervention behavior.

We further exclude three types of users likely to bias the estimates: membership users, third-party resellers, and bot-like accounts. Membership users who pay a monthly premium receive

Figure 2 **Timeline of Our Data Samples.**



free shipping regardless of cart value and are therefore unaffected by the CFS threshold policy. Third-party resellers who place large, frequent orders regardless of shipping policies are unlikely to respond to the threshold change. Bot-like users, identified by high-volume browsing behavior over long, continuous periods without corresponding transactions, likely do not represent real shopping activity. After filtering, the final sample comprises 261 consumers (132 from treated regions and 129 from control regions) who placed 1025 orders during the study period. To examine monthly purchase behavior, we construct an imbalanced panel dataset with 5,148 consumer-month observations, where each observation corresponds to a specific consumer in a given month. The panel is imbalanced because not all consumers were active at the beginning of the observation window.

3.2.1. Transaction data The transaction data provides detailed records of each completed order. Each record includes the consumer ID, the shipping address ZIP code, the product ID, timestamps for when each product was added to the cart, the quantity and unit price for each product, and the payment timestamp. The data also include information on the retailer's fulfillment costs.

To study product heterogeneity, we classify products by their popularity before the intervention. Specifically, we define popular products as those that collectively account for the top 80% of total units sold during the pre-intervention period. The remaining products are categorized as long-tail products, which together account for the bottom 20% of total units sold.

3.2.2. Clickstream data We obtained detailed consumer clickstream data that captures browsing behavior. Each log entry includes a timestamp, the consumer ID, and the ID of the viewed product.

Combining the transaction and clickstream data, we construct session-based consumer browsing activity. A session is defined as a continuous browsing period initiated when a consumer enters the website and terminated either by a transaction, a period of inactivity (e.g., 30 mins), or an exit (Cooley et al. 1999). Within each session, consumers may perform a variety of observable actions, including viewing product detail pages, adding items to the cart, and proceeding to checkout.

Table 1 Variable statistics

Variables	Observations	Mean	Standard deviation	Min	Max
$Spending_{it}$	5148	29.69	78.19	0	896.67
$AvgCartValue_{it}$	5148	27.06	69.25	0	636.36
$Transactions_{it}$	5148	0.20	0.45	0	5
$Variety_{it}$	5148	1.16	3.33	0	38
$AvgPrice_{it}$	5148	5.60	21.40	0	299
$PopShare_{it}$	5148	0.12	0.29	0	1

3.2.3. Variables Following prior research (e.g., Guo and Liu 2023), we define two sets of outcome variables that capture (1) how much a consumer purchases and (2) what a consumer purchases. Table 1 summarizes the outcome variables used in the analysis along with their descriptive statistics. The unit of analysis in this study is the consumer-month level.

To measure how much a consumer purchases, we use consumer monthly spending (*Spending*), defined as the total value of completed orders in a given month. Since total spending can be decomposed into the number of transactions and average cart value, we also include *Transactions*, defined as the number of completed orders in a month (a measure of purchase frequency), and *AvgCartValue*, defined as the average cart value per order in that month.

To examine what a consumer purchases, we consider several variables related to cart composition. *Variety* captures the number of distinct products a consumer purchases in a month and serves as a proxy for the variety of purchases. *AvgPrice* measures the average unit price of the total purchased items in a month, with higher values indicating a tendency toward more expensive products. *PopShare* captures the share of spending allocated to purchasing popular products, with higher values indicating greater preferences for popular items.

As shown in Table 1, all outcome variables exhibit substantial right skewness and contain a nontrivial mass at zero. To address these distributional features, we apply a log-plus-one transformation. Specifically, for each outcome variable $y \in \{Spending, AvgCartValue, Transactions, Variety, AvgPrice\}$, we construct the transformed variable as $\log(y_{it}+1)$, which we use consistently throughout the regression analyses in this paper. The variable *PopShare* is not transformed, as it is a bounded proportion and already lies within a well-defined range.

3.3. Model-free Evidence

We first provide some analyses of the overall impact of CFS threshold increases by comparing changes in purchase behavior between the control and treatment groups before and after the intervention (Table 2). The simple comparison shows that consumers' monthly spending in the treatment group declined more than that in the control group after the CFS threshold increase

Table 2 Model-free Evidence: Descriptive Statistics of Mean Value.

	Control			Treatment		
Variable	**Pre**	**Post**	**Change**	**Pre**	**Post**	**Change**
Spending	33.48	33.13	-1.1%	31.68	21.82	-31.1%
AvgCartValue	31.43	30.70	-2.3%	28.13	19.25	-31.6%
Transactions	0.22	0.19	-12.9%	0.24	0.16	-32.5%
Variety	1.38	1.21	-11.9%	1.29	0.83	-36.0%
AvgPrice	5.78	5.48	-5.2%	6.32	5.01	-20.8%
PopShare	0.14	0.12	-15.1%	0.15	0.10	-35.1%

(−31.1% vs. −1.1%). We find larger reductions in both average cart value (−31.6% vs. −2.3%) and purchase frequency (−32.5% vs. −12.9%). Regarding what consumers purchase, we find that consumers in the treatment group tend to purchase fewer distinct products and prefer cheaper and long-tail products. To visualize the impact, we also show the time series of consumer monthly spending during the observation periods in Figure 3, and include the time series for other outcome variables in the Online Appendix EC.1.

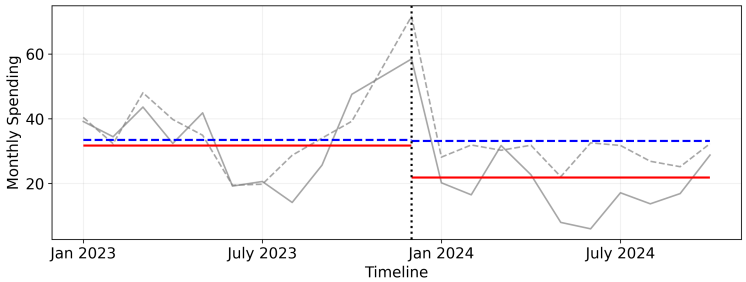
These model-free statistics provide preliminary evidence that raising the CFS threshold adversely affects consumer behavior, as measured by all outcome variables, in the treatment group relative to the control group. We further investigate these patterns in our empirical analysis by accounting for both observed and unobserved factors not captured in the above analysis.

4. Results

We employ a DiD approach with two-way fixed effects to estimate the causal impact on consumer purchase behavior. Our identification strategy relies on comparing treated and control consumers before and after the policy change, while controlling for time-invariant consumer heterogeneity and common temporal shocks. Specifically, we estimate the following model specification:

$$y_{it} = \alpha_0 + \alpha_1 \cdot Treated_i \times Post_t + \mu_i + \delta_t + \varepsilon_{it} \tag{1}$$

Figure 3 Time Series of Consumer Monthly Spending During the Observation Period



Notes: The grey dashed line represents the time series of the average monthly spending for the control group, and the grey solid line represents the same for the treatment group. The vertical dotted line marks the beginning of the post-intervention period. The blue-dashed horizontal lines indicate the pre- and post-intervention averages for the control group, and the red-solid horizontal lines indicate those for the treatment group.

Table 3 Effects of CFS Threshold Increase on Consumer Spending.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*
$Treated \times Post$	-0.216**	-0.211**	-0.032**
	(0.101)	(0.098)	(0.016)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.032	0.031	0.032

*Notes: Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

where y_{it} is one of the outcome variables described in Section 3.2.3 for consumer i in year-month t. $Treated_i$ is a binary variable equal to one if consumer i resides in a CFS-entitled region, and zero otherwise. $Post_t$ is a binary variable equal to one if month t is in the post-intervention period, and zero if in the pre-intervention period. The key explanatory variable is the interaction term $Treated_i \times Post_t$, and its coefficient α_1 captures the average treatment effect on the treated. The model includes consumer fixed effects μ_i to control for time-invariant heterogeneity across individuals, and year-month fixed effects δ_t to account for common time trends and seasonal fluctuations.

We also conduct aggregation-level analyses to understand the impact of the threshold increase on the platform's revenue and profitability. See Online Appendix EC.2 for the details.

4.1. Impact on Consumer Spending

Table 3 shows the regression results. Specifically, Column (1) reports the treatment effect on consumer monthly spending. The coefficient indicates that consumers in the treatment group substantially reduced their spending following the increase in the threshold. The coefficient suggests a 19.4% decline ($e^{-0.216} - 1 = -0.194$) in consumer spending in the treatment group after the intervention. To assess the economic significance, we use the post-intervention monthly spending of \$33.13 per consumer in the control group as the baseline. Relative to this benchmark, the threshold increase corresponds to a \$6.43 reduction in monthly spending per consumer in the treatment group. Column (2) presents the treatment effect on the average cart value per order. The estimated coefficient indicates that the average cart value declined by 19.0% in the treatment group relative to the control group. Column (3) shows the treatment effect on the number of completed orders. The coefficient suggests that, following the increase in the threshold, consumers in the treatment group placed approximately 3.1% fewer orders per month compared to those in the control group.

Our empirical results in Table 3 reveal nuanced dynamics in how consumers respond to increases in the CFS threshold. Prior empirical studies (e.g., Cachon et al. 2018, Hemmati et al. 2021, Lewis

Table 4 Effects of CFS Threshold Increase on Consumer Shopping Pattern.

	(1) *Variety*	(2) *AvgPrice*	(3) *PopShare*
$Treated \times Post$	-0.076*	-0.122*	-0.033**
	(0.040)	(0.062)	(0.015)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.022	0.038	0.037

*Notes: Standard errors are clustered at the consumer level. Both Variety and AvgPrice are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

et al. 2006) have shown that CFS policies are effective at consolidating orders and incentivizing consumers to purchase more items than unconditional or flat-fee shipping policies, resulting in higher cart values and higher overall revenue. However, we demonstrate that when the platform raises the CFS threshold, the incentive structure changes. Most notably, the decline in consumer monthly spending is driven primarily by a reduction in cart value rather than a substantial decline in purchase frequency. This pattern is consistent with consumers becoming less willing to expand their carts to reach the higher CFS threshold, ultimately eroding the effectiveness of the CFS incentive.

4.2. Impact on Consumer Shopping Pattern

Beyond our primary outcomes of spending, cart value, and purchase frequency, we conduct an exploratory analysis of changes in basket composition. Table 4 shows that, following the threshold increase, consumers purchased 7.3% fewer distinct products, the average unit price declined by 11.5%, and the share of spending allocated to popular products declined by 3.3 percentage points.

To sum up, the results in both Tables 3 and 4 reveal a consistent pattern: the increase in the CFS threshold not only suppressed how much consumers purchased, but also shifted the composition of what they purchased. Unlike the behavior documented in prior studies, consumers reduced both the average price and the variety of products purchased. These changes are consistent with a weakening of the cart-expansion incentives associated with CFS.

4.3. Robustness Checks

We assess the robustness of our main results by testing the parallel trends assumption and conducting placebo tests. We also address non-random assignment concerns using coarsened exact matching and mitigate potential selection bias with an inverse probability weighting estimator; see Online Appendices EC.7.4 and EC.7.5.

4.3.1. Parallel Trends Assumption. We examine the parallel trends assumption in DiD analysis, i.e., that the treatment and control groups follow parallel trends in the outcome variables

during the pre-intervention period. To do so, we conduct a pre-trends test by examining whether the differences in the outcomes between the treatment and control groups are statistically different from zero across the pre-intervention periods. We use the following model specifications for this test:

$$y_{it} = \tau_0 + \sum_m \tau_m \cdot Treated_i \times \mathbf{1}(t = m) + \mu_i + \delta_t + \varepsilon_{it} \tag{2}$$

where m represents a specific month in the pre-intervention period, and January 2023 serves as a reference month. This specification allows a different coefficient for each time point in the pre-intervention period. It does not impose a linear assumption about the trends in the differences between the treatment and control groups. We present the estimation results in Table EC.7 in the Online Appendix EC.7.1. We find that, across all outcome variables in Section 3.2.3, the interaction term coefficients are *insignificant*, suggesting no statistically significant differences in the trends of the dependent variables between the treatment and control groups during the pre-intervention period.

4.3.2. Placebo Tests We also conduct a placebo test to rule out the possibility that the estimated effect is driven by factors other than an increase in the CFS threshold. Specifically, we first introduce a "fake CFS threshold increase" during the pre-intervention period and estimate its impact on the dependent variables using the pre-intervention period data. We set our fake increase to be on June 1, 2023, which is the middle of the pre-intervention period in our sample. We use a specification that replaces the variable $Post_t$ in Specification (1) with the variable $PlaceboPost_t$, which is a dummy variable that equals one if time t is after the fake threshold increase and zero otherwise. Next, we introduce a "fake treatment group" among consumers in the actual control group and estimate the treatment effect on the dependent variables using the control group data. We randomly select 64 consumers from the control group and assign them to the fake treatment group, and the remaining 65 consumers stay in the control group. We use a specification that replaces the variable $Treated_i$ in Specification (1) with $PlaceboTreated_i$, a dummy variable that equals one if consumer i belongs to the placebo treatment group and zero otherwise. The estimation results are shown in Table EC.8 in the Online Appendix EC.7.2. We find no statistically significant effects of the placebo treatment on the dependent variables, suggesting that the estimated effects of the CFS threshold increase are not spurious.

5. Reference-Dependent Perceived Attainability

In this section, we examine why the increase in the CFS threshold reduces consumer spending and why the response is stronger for some consumers than for others. We organize the evidence around the attainability of the free-shipping goal. A CFS policy does more than attach a delivery fee to orders below a threshold: it creates a salient spending target. When that target appears attainable, consumers may add discretionary products or consolidate purchases to qualify for free shipping (Cachon et al. 2018, Chen and Ngwe 2018). In other words, consumers' shopping behavior and their baskets are adjusted due to the existence of the CFS thresholds. However, raising the threshold from \$80 to \$100 increases the distance to the reward. If the new target no longer appears worth pursuing, consumers may change their established shopping behaviors. Consequently, threshold-induced basket adjustment may disappear, resulting in a reduction in spending, average cart value, and transaction frequency.

This explanation yields two empirical implications. First, the adverse effect should be stronger among consumers whose pre-intervention behavior indicates that the original \$80 threshold had already shaped their baskets. Second, repeated and recent exposure to the original threshold should make it more likely to serve as a reference point, causing the new \$100 threshold to appear farther away and less attainable. To find supporting evidence for these implications, we estimate variants of the following triple-difference specification:

$$\begin{aligned} y_{it} = \beta_1(Treated_i \times Post_t) + \beta_2(Moderator_i \times Post_t) \\ + \beta_3(Treated_i \times Post_t \times Moderator_i) + \mu_i + \delta_t + \varepsilon_{it}, \end{aligned} \tag{3}$$

where each moderator serves as a proxy for the corresponding implication and is constructed exclusively from pre-intervention behavior.

5.1. Threshold-induced Adjustment Around the Original Threshold

The attainability mechanism requires that the original \$80 threshold was behaviorally consequential before the policy change. In particular, at least some consumers must have adjusted their baskets by adding products or consolidating purchases to qualify for free shipping. Final cart value alone provides only imperfect evidence of such adjustment: an order slightly above \$80 may reflect threshold-induced additions, but it may also reflect the consumer's natural demand in the absence of the CFS policy. We therefore use the sequence and timing of consumers' pre-intervention cart-building activities constructed with the clickstream data to identify observable patterns consistent with adjustment around the original threshold.

For each completed pre-intervention cart, we identify two forms of CFS-induced adjustment. The first is a *top-up pattern*. A cart is classified as exhibiting this pattern if, during the consumer's final shopping session, its running value enters the \$60–\$80 range, one or more subsequent additions raise the cart value to at least \$80, and the completed cart value does not exceed \$90. We select these cutoffs to capture economically plausible top-up behavior. Panel A of Figure 1 shows markedly fewer completed carts with values between \$60 and \$80, consistent with consumers whose carts fall within this range adding products to cross the free-shipping threshold. In addition, shipping fees in CFS-enabled regions range from \$11 to \$13, making cart additions of up to \$20 a plausible response to the free-shipping incentive when consumers also value the additional products.

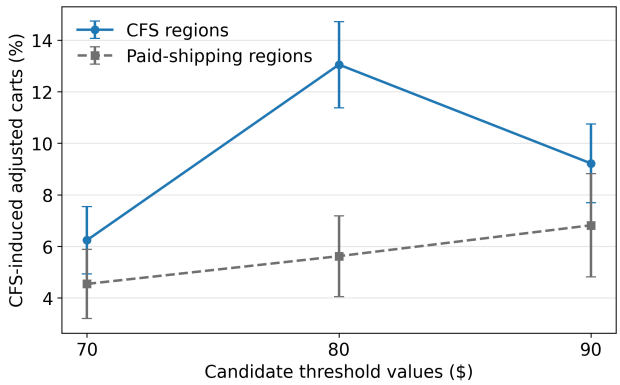
The second is a *consolidation pattern*. A cart is classified as exhibiting this pattern when products are added across at least two shopping sessions separated by at least one day. The products added during each session, if purchased separately, would generate a subtotal below \$80, whereas their combined value reaches or exceeds \$80. This pattern is consistent with consumers delaying or combining purchases to qualify for free shipping.

We distinguish these two patterns because they capture different conditions of adjustment to the CFS threshold: a top-up pattern reflects within-session, modest basket expansion. In contrast, the consolidation pattern reflects intertemporal order aggregation, which can produce larger baskets and fewer transactions when consumers combine purchases that would otherwise be made separately. Both patterns are well documented in the literature (e.g. Cachon et al. 2018, Lewis et al. 2006) and indicate that the original threshold altered the consumer's shopping behavior. We thus combine them when constructing prior CFS-induced behavior.

We first validate the cart classification using both threshold and regional placebos. We repeat the same classification at artificial thresholds of \$70 and \$90, shifting the top-up and consolidation windows accordingly. We also apply the procedure to consumers in paid-shipping regions, where no CFS threshold was present. Figure 4 shows that the prevalence of the identified adjustment patterns is substantially higher in CFS-enabled regions than in paid-shipping regions only at the actual \$80 threshold (13.05% for CFS-enabled region and 5.62% for Paid-shipping region, with $p < 0.001$). The concentration of the difference at the actual threshold implies that the classification captures behavior associated with the CFS incentive rather than routine additions near checkout, naturally large baskets, or order consolidation unrelated to free shipping.

Next, we obtain the propensity that a consumer's cart involves CFS-induced adjustment. For consumer i, let N_i denote the total number of completed pre-intervention carts. Let $A_{ij} = 1$ equal

Figure 4 Threshold and Regional Placebo Tests for Pre-Intervention Cart Adjustment



one if the completed pre-intervention cart j of consumer i exhibits either the top-up or consolidation pattern, and zero otherwise. We then define $A_i = \sum_{j=1}^{N_i} A_{ij}$ to represent the number of carts that exhibit at least one threshold-related adjustment pattern. The raw adjustment rate, A_i/N_i, provides a natural consumer-level measure. However, purchase histories are sparse and unequal across consumers. A raw rate based on one or two carts is therefore substantially less reliable than the same rate based on a longer purchase history.

We address this measurement error using a beta-binomial empirical-Bayes model. Let π_i denote consumer i's latent probability that a completed cart exhibits threshold-related adjustment. Conditional on π_i, the number of identified carts follows a binomial distribution:

$$A_i \mid \pi_i, N_i \sim \text{Binomial}(N_i, \pi_i).$$

We allow the latent probabilities to vary across consumers according to a population-level Beta distribution, i.e., $\pi_i \sim \text{Beta}(a, b)$, where a and b are unknown parameters. We estimate those parameters using completed pre-intervention carts from *all* consumers in the CFS-enabled regions. We restrict the estimation to the pre-intervention period so that the measure is not affected by behavioral responses to the threshold increase. The calibration procedures are provided in Online Appendix EC.4.1.

Conditional on the estimated population parameters, the posterior distribution of π_i is

$$\pi_i \mid A_i, N_i, \hat{a}, \hat{b} \sim \text{Beta}\left(A_i + \hat{a}, N_i - A_i + \hat{b}\right).$$

Thus, the posterior mean is

$$\hat{\pi}_i = \frac{A_i + \hat{a}}{N_i + \hat{a} + \hat{b}}. \tag{4}$$

Table 5 Heterogeneous Effects of the CFS Threshold Increase by Pre-Intervention Adjustment Propensity

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.232***	-0.225***	-0.029***
	(0.070)	(0.067)	(0.011)
$Treated \times Post \times Propensity$	-0.204**	-0.210**	-0.025**
	(0.101)	(0.097)	(0.012)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.032	0.031	0.032

*Notes: Standard errors are clustered at the consumer level. Outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

which can equivalently be written as

$$\hat{\pi}_i = \frac{N_i}{N_i + \hat{a} + \hat{b}} \frac{A_i}{N_i} + \frac{\hat{a} + \hat{b}}{N_i + \hat{a} + \hat{b}} \frac{\hat{a}}{\hat{a} + \hat{b}}.$$

Thus, $\hat{\pi}_i$ is the weighted average of the consumer's observed adjustment rate and the estimated population-average adjustment rate. For consumers with many completed carts, the posterior mean places greater weight on their own observed behavior. For consumers with sparse histories, it shrinks the noisy raw rate more strongly toward the population average. This construction preserves consumer heterogeneity while limiting the influence of isolated cart-building patterns.

We then use the standardized posterior mean as the moderator $Propensity_i$ in Specification (3) and report the results in Table 5. The coefficient on $Treated \times Post \times Propensity$ is negative for spending (-0.204, $p = 0.044$), average cart value (-0.210, $p = 0.031$), and transactions (-0.025, $p = 0.041$). Thus, the adverse effect of the threshold increase becomes progressively larger among consumers whose pre-intervention carts were more likely to exhibit threshold-related adjustment. This analysis provides evidence for the behavioral premise that the overall decline is stronger among consumers whose pre-intervention behavior was more likely to have been shaped by the original threshold.

5.2. Reference Points and Perceived Attainability

We next examine why the higher threshold discourages consumers from spending more on the platform. Reference-dependence theory suggests that consumers evaluate outcomes relative to benchmarks formed through prior experience rather than only in absolute terms (Kahneman and Tversky 1979, Tereyağoğlu et al. 2018). In our setting, repeated shopping under the original CFS policy may help establish the original threshold as a familiar benchmark for qualifying for free shipping. Goal-gradient research further suggests that effort toward a reward depends on the perceived distance to the target and, more generally, on whether the target is attainable and activated

(Choudhary et al. 2021, Kivetz et al. 2006). Once the original $80 threshold is encoded as a reference point, the new $100 threshold is then evaluated not simply as an absolute target but as a $20 increase over the familiar qualification level. This reference-dependent increase in subjective distance can reduce perceived attainability and lead consumers to abandon the CFS-induced behavior.

This logic separates two related conditions. First, repeated shopping under the original policy may help establish the $80 threshold as a reference point, a process we refer to as *reference-point formation*. Second, *reference-point recency* concerns how recently consumers experienced the original policy, which may influence how readily they recall this reference point when the threshold changes. We examine these two aspects using measures of pre-intervention purchase frequency and recency.

5.2.1. Reference-point formation. We use the frequency with which a consumer encountered the original threshold before the intervention as a proxy for reference point formation. Let N_i denote consumer i's number of completed orders during the pre-intervention periods and let T_i denote the consumer's observable pre-intervention tenure, measured in months. The raw rate, N_i/T_i, measures the observed monthly frequency of encounters with the original CFS policy. As with the raw adjustment rate used to construct $Propensity_i$ in the previous section, however, this rate is especially noisy for consumers with short or sparse histories.

We therefore use a Gamma–Poisson empirical-Bayes model. Conditional on consumer i's latent monthly order-arrival rate λ_i, the number of completed orders follows a Poisson distribution:

$$N_i \mid \lambda_i, T_i \sim \text{Poisson}(\lambda_i T_i).$$

We further assume that the latent arrival rates follow a population-level Gamma distribution, $\lambda_i \sim \text{Gamma}(r, \alpha)$, where the population-average monthly arrival rate is r/α. Integrating over the Gamma heterogeneity yields a negative-binomial distribution for consumers' pre-intervention order counts. We estimate r and α using the pre-intervention histories of consumers from both CFS-enabled and paid-shipping locations. Details of the population-level estimation are provided in Online Appendix EC.4.2.

Conditional on the estimated population parameters, the posterior distribution of the arrival rate is

$$\lambda_i \mid N_i, T_i, \hat{r}, \hat{\alpha} \sim \text{Gamma}\left(\hat{r} + N_i,\ \hat{\alpha} + T_i\right),$$

with posterior mean

$$\hat{\lambda}_i = \frac{\hat{r} + N_i}{\hat{\alpha} + T_i}. \tag{5}$$

which can be expressed as a weighted average:

$$\hat{\lambda}_i = \frac{T_i}{T_i + \hat{\alpha}} \frac{N_i}{T_i} + \frac{\hat{\alpha}}{T_i + \hat{\alpha}} \frac{\hat{r}}{\hat{\alpha}}.$$

Consumers with longer observable tenure therefore receive greater weight on their own observed order-arrival rate, whereas consumers with limited histories are shrunk more strongly toward the population means. This posterior mean arrival rate ensures that differences in estimated reference-point formation are not driven mechanically by the number of months for which a consumer is observed or by a single isolated order.

For estimation, we define $Formation_i$ as the standardized logarithm of the posterior arrival rate:

$$Formation_i = \frac{\log(\hat{\lambda}_i) - \overline{\log(\hat{\lambda})}}{s_{\log(\hat{\lambda})}}.$$

Higher values therefore indicate more frequent encounters with the platform and, for consumers in CFS-enabled regions, more opportunities for the original \$80 threshold to become encoded as the normal free-shipping requirement. We use $Formation_i$ as the moderator in Specification (3). Table 6 shows the results. The coefficients on $Treated \times Post \times Formation$ are negative and statistically significant at the 1% level for all outcome variables, implying that the threshold increase produces a larger decline among consumers who encountered the original policy more frequently, as expected when repeated exposure strengthens reference-point formation.

Table 6 Heterogeneous Effects of the CFS Threshold Increase by Reference-Point Formation

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.158*	-0.155*	-0.023*
	(0.085)	(0.083)	(0.013)
$Treated \times Post \times Formation$	-0.479***	-0.458***	-0.079***
	(0.063)	(0.064)	(0.008)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.039	0.038	0.042

*Notes: Standard errors are clustered at the consumer level. Outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

As robustness checks, we replace the empirical-Bayes posterior rate with the raw arrival rate N_i/T_i. We also use $Exposure_i$ as a proxy for reference-point formation. $Exposure$ represents the standardized logarithm of the total number of completed pre-intervention orders and captures cumulative shopping experience regardless of its timing. The results using these alternative measures are reported in Online Appendix EC.7.3 and yield consistent results as $Formation$.

Table 7 Heterogeneous Effects of the CFS Threshold Increase by Reference-Point Recency

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	0.073	0.075	0.013
	(0.105)	(0.104)	(0.015)
$Treated \times Post \times Recent$	-0.550***	-0.543***	-0.085***
	(0.192)	(0.187)	(0.030)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.034	0.034	0.035

*Notes: Standard errors are clustered at the consumer-level. Outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

5.2.2. Reference-point recency. Frequency of prior exposure captures whether the original threshold was likely to have been formed, but an established reference point may be less influential when the consumer has not encountered it recently. We therefore use the time between the consumer's final pre-intervention completed order and the policy change as a proxy for reference-point recency. Specifically, $Recent_i$ equals one if the consumer's final pre-intervention order occurred within three months of the policy change, which is the sample-median interval, and zero otherwise. As shown in Table 7, the coefficient on $Treated \times Post \times Recent$ is negative and statistically significant for all three outcomes. We do not detect a statistically significant effect on consumers who do not have recent purchases. In contrast, consumers with recent purchases significantly reduce their spending ($0.073 - 0.550 = -0.477$, $p = 0.003$), average cart value ($0.075 - 0.543 = -0.468$, $p = 0.003$) and transactions ($0.013 - 0.085 = -0.072$, $p = 0.005$). Thus, consumers who purchased more recently under the original policy exhibit a stronger adverse response to the threshold increase.

5.2.3. Short- and long-term effects. We further investigate whether the reference-dependent perceived attainability exhibits a temporal effect. Specifically, we categorize the post-intervention periods into three groups: an Early period from December 2023–February 2024, an Intermediate period from March–July 2024, and a Late period from August–October 2024. We replace the single $Treated \times Post$ interaction with separate interactions in both the DiD and triple DiD analyses. Although the point estimates are less negative for Late, the differences are not statistically significant. As a result, we do not detect temporal differences within the observed period. We provide detailed analyses in Online Appendix EC.3.

5.3. Experimental Evidence on Reference-Dependent Perceived Attainability

We conduct a randomized online controlled experiment on Prolific to test whether described prior-threshold history affects perceived attainability and willingness to pursue free shipping while holding current purchase requirements constant. Participants were randomly assigned to a prior-\$80 threshold condition (*Prior80*) or a prior-\$100 threshold condition (*Prior100*). After excluding

participants who failed attention checks or did not complete the experiment, the final sample comprised $N = 160$ participants: 81 in *Prior80* and 79 in *Prior100*.

Participants were asked to imagine that they had shopped regularly at FreshMarket Online, a hypothetical online grocery platform, during the previous year, when the free-shipping threshold was \$80 or \$100, depending on their assigned condition. They were then informed that today's threshold was \$100 and viewed a current grocery cart totaling \$85, a subtotal selected through pretesting to avoid floor or ceiling effects in the choice to continue shopping. Participants selected whether to (1) continue shopping and then check out with free shipping; (2) pay the \$5.99 shipping fee and check out with the current cart; or (3) leave the cart without checking out at that time. The current threshold, cart subtotal, additional spending requirement, and shipping fee were identical across conditions; only the described prior-threshold history differed.

After making their checkout choice, participants completed a questionnaire assessing their perceptions of the shopping situation and policy change and provided demographic information. Following Lembregts and Pena-Marin (2021), Zhang and Huang (2010), we measured perceived attainability using the single item, "Qualifying for free shipping on today's order felt attainable." We also measured CFS-induced motivation using the item, "I felt motivated to add items to my cart to qualify for free shipping," following Lembregts and Pena-Marin (2021). Both items used seven-point agreement scales (1 = strongly disagree; 7 = strongly agree).

All participants resided in the United States. 52.5% reported being female, 67.5% identified as White, and 70.6% reported full- or part-time employment. Mean age was 42.5 years ($SD = 12.2$). Comparisons of the measured demographic characteristics did not detect statistically significant differences between conditions.

Our manipulation check asked whether today's free-shipping threshold felt like a substantial increase from the previous threshold, using a seven-point agreement scale. Participants in *Prior80* reported a larger perceived increase than those in *Prior100* ($M = 5.20$ versus 2.53; $p < 0.001$), indicating that the manipulation changed perceptions of the policy increase in the intended direction.

We first examined the distribution of the three checkout choices. In *Prior80*, 60.5% chose to continue shopping to qualify for free shipping, 16.0% selected checkout with paid shipping, and 23.5% selected leaving the cart without checking out. The corresponding percentages in *Prior100* were 79.7%, 12.7%, and 7.6%. A Pearson chi-square test indicated that the overall choice distribution differed between conditions ($\chi^2(2) = 8.878$, $p = 0.012$). We then estimated a binary logistic regression, coding the choice to continue shopping to qualify for free shipping as one and

either alternative as zero. Participants in *Prior80* had lower odds of selecting this option than those in *Prior100* after controlling for demographic characteristics (odds ratio = 0.358, $p = 0.0065$). Regarding the perceptions of the shopping situation, participants in *Prior80* perceived the free-shipping threshold as less attainable than those in *Prior100* ($M = 5.07$ versus 5.81; $p = 0.0027$). CFS-induced motivation was 4.48 in *Prior80* and 5.57 in *Prior100* ($p = 0.0005$).

Together, the choice and attainability results provide experimental evidence that assigned prior-threshold history influences responses to identical current checkout requirements. Participants in *Prior80* were less likely to choose to continue shopping to qualify for free shipping and perceived qualification as less attainable than those in *Prior100*. These results are consistent with the proposed reference-dependent perceived attainability mechanism.

6. Optimal Threshold Design

In this section, we address the last research question: What does the history-dependent response imply for revising an established CFS policy? We focus on a single-cart framework and examine how to set the free shipping threshold to maximize consumer net revenue at the cart level. There are two reasons for us to adopt such a framework: First, the single-cart setting allows us to explicitly model both the shipping surcharge paid by the consumer and the fulfillment cost incurred by the retailer, which are central to threshold design. Second, the threshold increase has only a marginal effect on purchase frequency, implying that changes in cart value are the primary driver of revenue outcomes.

Our counterfactual analysis proceeds as follows. We first analyze the optimal threshold under the assumption of homogeneous consumers, using the estimated elasticity of cart value with respect to the threshold. We then relax this homogeneity assumption by explicitly accounting for heterogeneity in cart values. Next, we discuss how the platform can operationalize threshold increases in practice by leveraging the reference dependent perceived attainability.

6.1. The Optimal Thresholds

We assume that the cart value, denoted by $y(\tau)$, responds with a constant elasticity to the free shipping threshold τ locally around the observed policy change. Specifically, we assume a log–log relationship between cart value and the threshold, i.e., $\ln y(\tau) = \ln y(\underline{\tau}) + \hat{c}\, \ln\left(\tau/\underline{\tau}\right)$, which implies

$$y(\tau) = y(\underline{\tau}) \cdot \left(\tau/\underline{\tau}\right)^{\hat{c}}. \tag{6}$$

Here, $\underline{\tau}$ marks the original threshold value, $y(\underline{\tau})$ is a parameter representing the observed average cart value at the original threshold, and $\hat{c}$ is the elasticity parameter calibrated from the observed DiD estimation when the threshold increases from $\underline{\tau}$ to $\overline{\tau}$.

The platform incurs a per-order fulfillment cost of C dollars and may charge the consumer a shipping surcharge of $S(y,\tau) \geq 0$ dollars, which depends on the realized cart value y and the free-shipping threshold τ. We model the shipping surcharge as

$$S(y,\tau) = \begin{cases} C - ky & y < \tau \\ 0 & y \geq \tau \end{cases}$$

where $k \in [0, C/\tau]$ governs how the surcharge varies with cart value. When $k = 0$, the policy corresponds to a standard CFS policy with a fixed surcharge below the threshold. When $k > 0$, the surcharge declines with cart value, capturing a partially discounted CFS policy. $k \leq C/\tau$ guarantees that the surcharge is still positive before the cart value hits the threshold. The platform chooses the free-shipping threshold within the range $[\underline{\tau}, \overline{\tau}]$ to maximize its expected per-order net revenue:

$$\tau^* \in \arg\max_{\underline{\tau} \leq \tau \leq \overline{\tau}} \; y(\tau) + S(y(\tau), \tau) - C$$

Consequently, we have the following result to characterize the optimal threshold:

PROPOSITION 1 **(Optimal Free Shipping Thresholds)**. *Let $c < 0$. Define $\hat{\tau}(a) = \left(a \cdot \underline{\tau}^{-c}\right)^{\frac{1}{1-c}}$. Then, there exists a cutoff $a^* > 0$, implicitly defined by $(1-k)\hat{\tau}(a^*) = a^* - C$, such that the optimal free-shipping threshold satisfies $\tau^* = \hat{\tau}(a)$ if and only if $a \in [\underline{\tau}, a^*]$ and $\hat{\tau}(a) \leq \overline{\tau}$; otherwise, $\tau^* = \underline{\tau}$.*

The proof can be seen in Online Appendix EC.6.1. The parameter a can be interpreted as the observed average cart value at the baseline free-shipping threshold $\underline{\tau}$. Proposition 1 shows that the platform should increase the free-shipping threshold only when this original demand level is sufficiently close to the original threshold.

The key intuition is that the platform's profit is strictly decreasing in the threshold τ within both regimes, whether orders qualify for free shipping or incur a shipping surcharge. As a result, the optimal policy must lie at one of two points: the minimum allowable threshold $\underline{\tau}$, or the kink $\hat{\tau}(a)$ at which the representative order is just excluded from free shipping. Raising the threshold beyond this kink only reduces demand without generating additional shipping revenue. The cutoff a^* is determined by the indifference condition $(1-k)\hat{\tau}(a^*) = a^* - C$, which equates the profit from extracting shipping revenue at the kink to the profit from offering free shipping at the original

threshold. When $a \le a^*$ and the kink is feasible, increasing the threshold to $\hat{\tau}(a)$ allows the platform to monetize shipping without inducing a large reduction in the cart value. When original cart values are higher, however, the loss in demand from raising the threshold outweighs the shipping revenue that can be extracted, making it optimal to keep the threshold at its minimum level $\underline{\tau}$.

To operationalize the model, we calibrate the elasticity parameter using the log change in average cart value reported in Table 3. Specifically, the estimated elasticity is $\hat{c} \approx \frac{-0.211}{\log(100/80)} = -0.946$. We then numerically search for the optimal threshold. We consider original average cart values $y(80)$ ranging from \$70 to \$130. Under the log–log assumption, we set $\underline{\tau} = 80$ and $\overline{\tau} = 100$. We fix the per-order fulfillment cost at $C = 12$ and set $k = 0$. With the above parameters, we can obtain $a^* = 103.2$.

Figure 5 Optimal Threshold Values with Various Base Cart Values

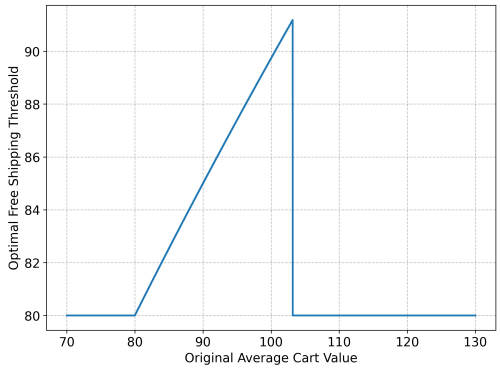


Figure 5 presents the counterfactual results. It follows Proposition 1 and shows a pronounced nonlinear relationship between the optimal thresholds and the original average cart value $y(80)$. Specifically, when $y(80)$ lies between \$80 and \$103.2, a moderate increase in the threshold becomes optimal. In this range, although a slightly higher threshold discourages consumer spending, the platform benefits from reducing subsidies for free shipping and collecting additional shipping surcharges. In contrast, when $y(80)$ is larger than \$103.2, it is optimal not to raise the threshold. The reason is that the cart value reduction dominates, and the loss cannot be offset by additional shipping surcharges. For our partner platform, whose pre-treatment average cart value was \$130.66 for CFS-enabled consumers, it should not increase the threshold at all.

We vary the discounting factor k and report additional insights in Online Appendix EC.6.2. Without accounting for the potential impact of shipping discounts on cart value, our results show that offering larger discounts on shipping surcharges makes suboptimal threshold choices more

costly. Given that determining optimal thresholds is far simpler, our results suggest the platform should not offer partially discounted CFS policies.

We also extend the preceding analysis to a stochastic threshold optimization problem by allowing cart values at the original free-shipping threshold, $y(\underline{\tau})$, to be heterogeneous across consumers. Specifically, we assume that baseline cart values follow a uniform distribution and study how the platform optimally chooses the free-shipping threshold to maximize expected per-cart net revenue.

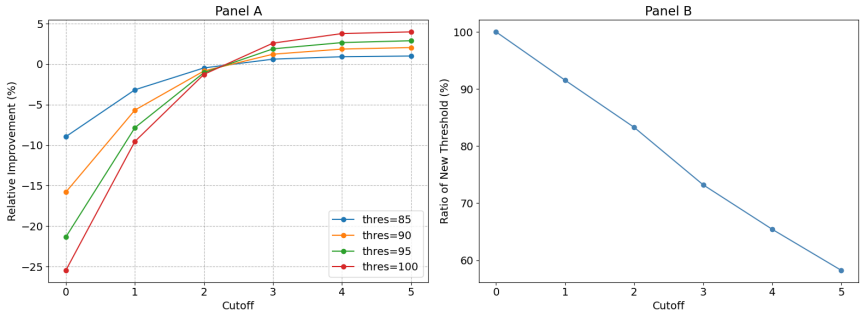
Although stylized, this uniform specification delivers a clear economic insight: the optimal free-shipping threshold depends not only on the average but also on the dispersion of cart values. Increasing the threshold τ raises the qualification cutoff and shifts a subset of orders from the free-shipping regime into the surcharge regime. Unlike the deterministic case, the marginal benefits therefore depend on the amount of probability mass near this cutoff, that is, on the distribution of baseline cart values rather than on their mean alone. In particular, holding the average cart value fixed, greater dispersion reduces the density of orders around the cutoff and weakens the incentive to raise the free-shipping threshold. We formalize these results in Online Appendix EC.6.3, where we derive the expected profit function and characterize the optimal threshold in Proposition EC.1.

6.2. Staggered Threshold Increases

For platforms that must raise their established free-shipping threshold, we examine a staggered, one-time transition based on consumers' purchase histories. The platform selects a single final threshold and a cutoff for the elapsed time since a consumer's last completed purchase. Each consumer retains the original threshold until this interval exceeds the cutoff. At the beginning of the consumer's next shopping session, the platform permanently assigns the account to the new threshold, which remains unchanged throughout that session and subsequent visits. Such a strategy may mitigate the adverse effect due to reference point while still helping the platform address rising fulfillment costs.

To evaluate the potential benefits of staggered implementation via selective exposure, we use a Monte Carlo simulation with bootstrapped data generation. We characterized consumers by the number of orders completed before the policy change and estimated the empirical distribution. Then, we randomly assign each simulated consumer a type by sampling from the empirical distribution. After that, we generate a sequence of purchase events over time using a bootstrapping procedure. Specifically, inter-purchase times and baseline cart values are resampled from pre-treatment transactions for consumers of the same type. This approach ensures that the simulated purchase process matches observed heterogeneity in purchase frequency and spending behavior.

Figure 6 Simulated Profit Improvement and Adoption Under Staggered Rollout.



Notes: The final adoption rate is independent of the threshold values. Therefore, the right panel shows the adoption rate as a function of the cutoff value with the threshold of $100.

Under the staggered rollout policy, the platform displays a higher threshold only to consumers whose current inter-purchase time exceeds a predetermined cutoff. Once a consumer is exposed, the higher threshold remains in effect for all subsequent purchases. Upon exposure, the consumer's realized cart value is adjusted to reflect the reference dependence effect using (6), with the parameter $\hat{c}$ determined as follows:

$$\hat{c} = \frac{\hat{\beta}_1 + \hat{\beta}_2 \log(\text{InterTime})}{\log(100/80)}$$

where InterTime denotes the consumer's current inter-purchase time, $\hat{\beta}_1$ captures the baseline impact of an increased threshold, and $\hat{\beta}_2$ captures the differential impact moderated by the strength of reference recency. Using a Triple DiD analysis with continuous inter-purchase time in Online Appendix EC.6.4, we obtain $\hat{\beta}_1 = -0.674$ and $\hat{\beta}_2 = 0.380$.

We track two key outcomes. First, we compute the platform's profit under a staggered rollout and compare it with that under the original threshold. We use the relative profit improvement to quantify potential profitability gains. Second, we monitor the proportion of consumers who see the new threshold at the end of the simulation horizon, which captures the speed of adoption of the higher threshold under the gradual rollout policy. We set the number of consumers to 132 (reflecting the retained consumers in the treatment group) and the simulation horizon to 12 months. We consider several final threshold alternatives ($85, $90, $95, and $100) and interval cutoff alternatives (from 0 to 5 months). Note that a cutoff of 0 corresponds to immediately exposing the higher threshold to all consumers, mirroring the policy implemented by our partner platform.

Figure 6 reports the simulated outcomes. The left panel shows that a staggered rollout strategy can substantially mitigate or even fully eliminate the profit losses associated with higher thresholds among retained consumers. By selectively exposing the higher threshold to consumers who are less affected by the reference point, the platform dampens the negative response that arises under

uniform implementation. In particular, when the platform restricts exposure to consumers with inter-purchase times exceeding 3 months, the staggered rollout policy yields higher profits. Beyond this point, further increases in the cutoff yield additional profit gains, though these gains exhibit diminishing marginal returns. In addition, if an appropriate cutoff value is selected, the new threshold is less important, since staggered rollout policies with different thresholds yield similar outcomes.

In contrast, the right panel highlights the corresponding trade-off in adoption speed. More selective rollouts necessarily slow the diffusion of the higher threshold across the consumer base. As the cutoff increases, a larger fraction of consumers continues to see the original threshold at the end of the simulation horizon. For example, with a 5-month cutoff, only 58% of consumers are subject to the new threshold after one year. Thus, while higher cutoffs improve short-run profit performance by limiting exposure among consumers recently exposed to the old threshold, they also delay the full adoption of the new threshold and complicate platform operations. Taken together, moderate cutoffs, such as exposing the higher threshold only to consumers with inter-purchase times longer than 2 or 3 months, can deliver most of the profit benefits of a staggered rollout while avoiding excessively slow adoption.

7. Conclusion

In this paper, we empirically investigate the causal effect of increasing the CFS threshold on consumer purchase behavior using a DiD approach at the consumer-month level. We find that, among continuing purchasers, raising the CFS threshold significantly reduces consumer monthly spending, driven primarily by a lower average cart value, with a marginal decline in purchase frequency. Furthermore, our aggregation-level analysis indicates that raising an established CFS threshold may lead to sales loss and profit reduction.

To explain this heterogeneity in consumers' responses, we examine reference-dependent perceived attainability as a potential mechanism. We first show that consumers have varying probabilities of adjusting their carts under the original CFS threshold, and that adverse effects are stronger among consumers who are more likely to adjust. Drawing on reference dependence and goal-gradient theory, we propose that repeated shopping under the original CFS policy may help establish the original threshold as a benchmark for qualifying for free shipping. When the threshold increases, comparison with this benchmark can make the new requirement seem farther away and less attainable, weakening consumers' willingness to engage in CFS-induced cart adjustment.

We find evidence consistent with this mechanism: consumers who purchase more frequently and more recently before the intervention show larger adverse responses. We also complement the field evidence with a randomized online controlled experiment that provides additional evidence for the proposed mechanism.

These findings offer important implications for managers of online retail platforms. First, the observed average cart value under the original CFS policy could be a misleading signal for deciding whether a threshold increase is beneficial. The observed average cart value reflects consumers' CFS-induced behavior in response to the existing threshold rather than their underlying purchase needs. When the threshold increases, these behavioral responses change, weakening CFS policy incentives and potentially leading to lower realized cart values that cannot be compensated by shipping surcharges. Therefore, a higher threshold can be profitable only when the existing average cart value lies sufficiently close to the current threshold. Second, for platforms that must raise their thresholds and can maintain account-level policy assignments, we examine a staggered, permanent transition based on purchase recency. Our simulations suggest that this approach can mitigate losses relative to an immediate uniform increase, while delaying adoption of the new threshold.

We conclude by outlining the study's limitations and suggesting directions for future research. First, our study is based on limited data from a single online grocery platform without a return policy. Therefore, future research should examine similar threshold adjustments in other retail settings, such as different product types, customer demographics, cultural contexts, or marketplaces with return options, to identify the boundary conditions of our findings. Second, our focus is on behavior change within retained consumers. Future research may further investigate the effects of increased thresholds on new consumers and consumer churn. Third, our study covers only a limited post-intervention period and does not identify effects over time. It is unclear whether the observed behavior changes and the mechanism persist in the long run. Future research thus can extend the observation period to study persistence, learning, and habit formation. Fourth, we focused mainly on the reference-dependent perceived attainability mechanism and examined budget constraints as a complementary explanation in Online Appendix EC.5. Future research may look into other explanations. For example, Chen and Ngwe (2018) show that search costs matter in CFS-induced behavior, and it would be interesting to see whether they still play a role when the threshold changes.

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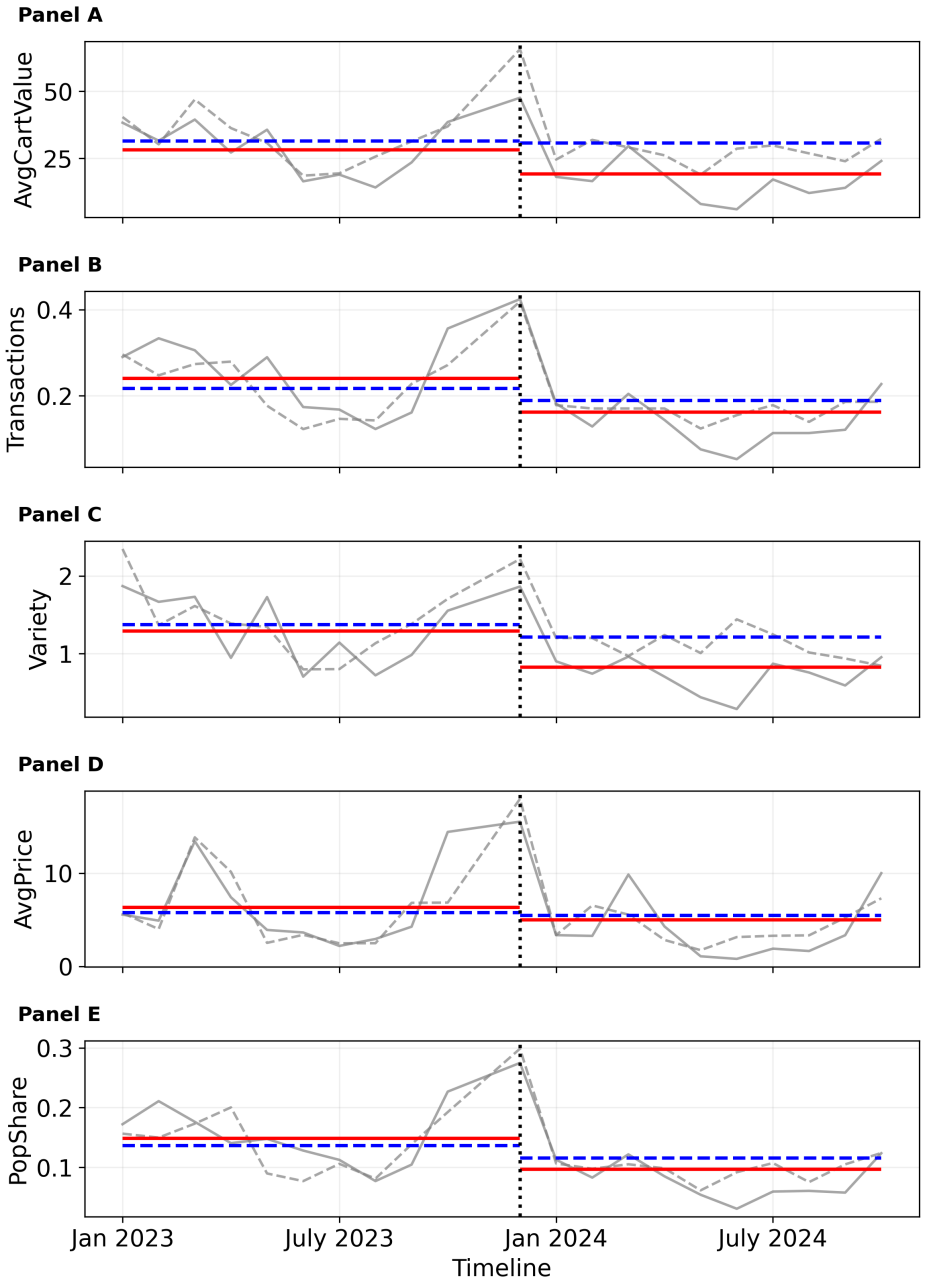
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Electronic Companion

EC.1. Model-free Evidence

Figure EC.1 Model-free Evidence: Time Series of Outcome Variables During the Observation Period



Notes: The grey dashed line represents the time series of the average outcome variable for the control group, and the grey solid line represents the same for the treatment group. The vertical dotted line marks the beginning of the post-intervention period. The blue dashed horizontal lines indicate the pre- and post-intervention averages for the control group, and the red solid horizontal lines indicate those for the treatment group.

EC.2. Aggregation Level Analyses

To investigate the effect of the threshold increase on the platform's revenue and profitability, we conduct an aggregated analysis using a balanced panel at the borough-month level. Similar to the consumer-level analyses in the main paper, we exclude three types of users: membership users, third-party resellers, and bot-like accounts. We then include the remaining consumers and their purchases, and aggregate them to the borough level based on their ZIP code. Our data includes 302 boroughs and 21 months, resulting in a balanced panel with 6,342 observations. We highlight that the aggregation not only includes every pre-intervention consumer regardless of their post-intervention activities, but also new consumers who arrived only after the threshold increase.

Specifically, we consider the following outcome variables: (1) $Sales_{jt}$ denotes the total value of completed orders from all consumers in borough j during year-month t; it captures the changes in the platform's overall revenue. (2) $AvgCartValue_{jt}$ captures the average order size within a borough. (3) $Transactions_{jt}$ represents the total number of completed orders. (4) $TransPayShip_{jt}$ represents the number of completed orders without free shipping. (5) $NetRev_{jt}$ indicates the net revenue by excluding platform-side shipping costs. It serves as a proxy for profitability and is calculated as follows:

$$NetRev_{jt} = \sum_{i=1}^{N_{jt}} (CartValue_i + ShippingSurcharge_i - ShippingCost_i)$$

where N_{jt} is the number of completed carts in borough j in year-month t. For each completed order i, consumer has to pay the cart value and the shipping surcharge (if any), the platform then pays a shipping cost for delivery. Every outcome variable y_{jt} is transformed by $\log(y_{jt}+1)$.

We use a difference-in-differences with two way fixed effects specification:

$$y_{jt} = \alpha_0 + \alpha_1 \cdot Treated_j \times Post_t + \mu_j + \delta_t + \varepsilon_{jt} \tag{EC.1}$$

where $Treated_j = 1$ if borough j is subject to the CFS policy and 0 otherwise. μ_j controls for borough-specific unobserved heterogeneity.

Table EC.1 reports the results. The coefficient for aggregate sales is -0.316 and statistically significant at the 10% level, which corresponds to a 27.1% reduction in monthly borough-level sales. The coefficient for average cart value is -0.349 and statistically significant at the 5% level. The overall transaction-count coefficient is negative but statistically insignificant, whereas the transaction-count with paid shipping coefficient is positive (0.048, $p = 0.149$) but not statistically

Table EC.1 Effects of Threshold Increase: Borough-Month Analysis with All Transactions

	Sales	*AvgCartValue*	*Transactions*	*TransPayShip*	*NetRev*
$Treated \times Post$	-0.316*	-0.349**	-0.021	0.048	-0.316*
	(0.178)	(0.162)	(0.039)	(0.034)	(0.177)
Borough fixed effects	Yes	Yes	Yes	Yes	Yes
Yearmonth fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	6,342	6,342	6,342	6,342	6,342
Adjusted R^2	0.393	0.317	0.596	0.472	0.395

*Notes: Standard errors are clustered at the borough level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

significant. Regarding the net revenue, the threshold increase has a marginally significant adverse effect at the 10% level.

The estimates show that increasing the threshold leads to revenue reduction, which mainly results from the reduction in the average cart value rather than the shopping frequency (the number of transactions drops, but the coefficient is not statistically significant). The platform increased the threshold in order to share the shipping costs with more consumers. The intervention seems to be doing its work, as the number of orders with consumer-paid shipping fees rises, but it is statistically insignificant. However, the shipping surcharge paid by consumers does not offset their spending reduction, resulting in a decrease in net revenue.

EC.3. Temporal Treatment Effects

We examine whether the purchasing response to the threshold increase changes over the eleven full post-intervention months available in our data. We distinguish an *Early* period, December 2023–February 2024; an *Intermediate* period, March–July 2024; and a *Late* period, August–October 2024. The early and late periods each contain three months, while the intermediate period contains five months. The choice of three-month windows is informed by the pre-intervention distribution of consumer inter-arrival times, which has a median of 3 months and a mean of 3.7 months.

We first replace the single $Treated_i \times Post_t$ interaction in the main DiD specification with separate interactions for the three post-intervention periods:

$$\begin{aligned} y_{it} = \beta_1(Treated_i \times Early_t) + \beta_2(Treated_i \times Intermediate_t) \\ + \beta_3(Treated_i \times Late_t) + \mu_i + \lambda_t + \varepsilon_{it}, \end{aligned} \tag{EC.2}$$

The coefficients β_1, β_2, and β_3 capture the period-specific treatment effects relative to the pre-intervention period. In particular, $\beta_3 - \beta_1$ measures the change in the treatment effect between the early and late periods.

As reported in Table EC.2, all coefficients are negative. However, only the Intermediate coefficients are significant at the 10% level. The difference between the Late effect and the Early effect

Table EC.2 Effects Across Early, Intermediate, and Late Post-intervention Periods

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Early$	-0.219	-0.218	-0.028
	(0.163)	(0.160)	(0.024)
$Treated \times Intermediate$	-0.234^{*}	-0.220^{*}	-0.039^{*}
	(0.131)	(0.128)	(0.020)
$Treated \times Late$	-0.183	-0.189	-0.024
	(0.147)	(0.144)	(0.023)
Late − Early	0.036	0.029	0.004
	(0.207)	(0.203)	(0.031)
p-value	0.862	0.887	0.897
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.031	0.031	0.032

*Notes: Early is December 2023–February 2024; Intermediate is March 2024 – July 2024; Late is August–October 2024. Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

is 0.036 ($p = 0.862$) for spending, 0.029 ($p = 0.887$) for average cart value, and 0.004 ($p = 0.897$) for transactions. Although the point estimates are less negative for Late, the differences are not statistically significant. Moreover, joint Wald tests of the null hypothesis that the three period-specific coefficients are equal ($\mathcal{H}_0 : \beta_1 = \beta_2 = \beta_3$) yield p-values of 0.942, 0.976, and 0.762, respectively, for all three outcomes. Thus, we do not detect statistically significant temporal differences within the observed period.

Table EC.3 Effects by Post-intervention Periods and Pre-intervention Purchases Recency

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Early$	0.073	0.064	0.022
	(0.232)	(0.229)	(0.034)
$Treated \times Intermediate$	0.005	0.022	-0.002
	(0.162)	(0.158)	(0.024)
$Treated \times Late$	0.188	0.176	0.031
	(0.188)	(0.184)	(0.029)
$Treated \times Early \times Recent$	-0.557^{*}	-0.537^{*}	-0.094^{**}
	(0.319)	(0.313)	(0.047)
$Treated \times Intermediate \times Recent$	-0.461^{*}	-0.464^{*}	-0.071^{*}
	(0.257)	(0.251)	(0.039)
$Treated \times Late \times Recent$	-0.690^{**}	-0.679^{**}	-0.102^{**}
	(0.289)	(0.283)	(0.045)
Late − Early (Recent = 1)	-0.019	-0.029	0.001
	(0.268)	(0.264)	(0.040)
p-value	0.945	0.912	0.987
Period $\times$*Recent* Interactions	Yes	Yes	Yes
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.035	0.034	0.035

*Notes: Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

Next, we examine whether the treatment-effect heterogeneity is associated with *Recent* changes over time. Specifically, we interact *Recent* with each period-specific treatment interaction in the

following triple DiD analysis.

$$\begin{aligned} y_{it} =& \beta_0 + \beta_1 \cdot Treated_i \times Early_t + \beta_2 Treated_i \times Intermediate_t + \beta_3 \cdot Treated_i \times Late_t \\ &+ \beta_4 Treated_i \times Early_t \times Recent_i + \beta_5 Treated_i \times Intermediate_t \times Recent_i \\ &+ \beta_6 Treated_i \times Late_t \times Recent_i + \beta_7 Early_t \times Recent_i \\ &+ \beta_8 Intermediate_t \times Recent_i + \beta_9 Late_t \times Recent_i + \mu_i + \delta_t + \varepsilon_{it} \end{aligned}$$

Table EC.3 reports the results. The triple-interaction point estimates are negative and significant at the 10% level for *Early* and *Intermediate*; it is negative and significant at the 5% level for *Late*. The late-minus-early differences for recent purchasers, $(\beta_3 + \beta_6) - (\beta_1 + \beta_4)$, are -0.019 ($p = 0.945$) for spending, -0.029 ($p = 0.912$) for average cart value, and 0.001 ($p = 0.987$) for transactions. None of these differences is statistically significant. Accordingly, this analysis does not provide precise evidence that recency-related treatment heterogeneity changes over the observation window. Thus, these findings do not establish whether the underlying mechanism itself weakens over time, nor do they show the mechanism has a constant effect over time.

EC.4. Estimating Population-level Propensity and Formation

This section describes how we estimate the population-level distributions used to construct the pre-intervention basket-adjustment propensity and reference-point formation measures. Both procedures use empirical Bayes: we first integrate out the unobserved consumer-level parameter and estimate the population parameters by marginal maximum likelihood, and then use the estimated population distribution to obtain each consumer's posterior distribution. Calibration uses only pre-intervention observations and is not restricted to consumers who purchase in both the pre- and post-intervention periods. Each consumer contributes one count-history record to the likelihood.

EC.4.1. Beta Distribution for CFS-induced Cart Adjustment Propensity

We follow the notation in Section 5.1. Specifically, for consumer i, let N_i be the number of pre-intervention carts and let A_i be the number exhibiting the CFS-induced cart adjustment pattern defined in the main text. Conditional on the observed history size, we specify

$$A_i \mid \pi_i, N_i \sim \text{Binomial}(N_i, \pi_i), \qquad \pi_i \sim \text{Beta}(a, b), \qquad a, b > 0,$$

where π_i is the latent probability that a completed cart exhibits the classified adjustment pattern. The population density is

$$f(\pi \mid a, b) = \frac{\pi^{a-1}(1-\pi)^{b-1}}{B(a, b)}, \qquad 0 < \pi < 1,$$

where $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the beta function. The population mean is $m = a/(a+b)$ and the concentration is $\kappa = a+b$, with variance $m(1-m)/(\kappa+1)$.

Because π_i is unobserved, we integrate it out rather than fit a Beta distribution directly to the noisy empirical rates A_i/N_i. The resulting beta-binomial probability is

$$\begin{aligned}\Pr(A_i \mid N_i, a, b) &= \int_0^1 \binom{N_i}{A_i} \pi^{A_i}(1-\pi)^{N_i - A_i} f(\pi \mid a, b)\, d\pi \\ &= \binom{N_i}{A_i} \frac{B(A_i + a, N_i - A_i + b)}{B(a,b)}.\end{aligned}$$

Assuming independence across consumer histories, the marginal log-likelihood is

$$\ell(a,b) = \sum_{i \in C} \left[\log \binom{N_i}{A_i} + \log B(A_i + a, N_i - A_i + b) - \log B(a,b) \right]. \tag{EC.3}$$

where C is the set of consumers in CFS-enabled regions with at least one pre-intervention purchase. The estimates satisfy

$$(\hat{a}, \hat{b}) \in \arg\max_{a>0,\, b>0} \ell(a,b).$$

We optimize over the transformed parameters

$$\eta = \log \frac{m}{1-m}, \qquad \xi = \log \kappa, \qquad a = \frac{e^\eta}{1+e^\eta} e^\xi, \qquad b = \frac{1}{1+e^\eta} e^\xi.$$

This parameterization enforces $a, b > 0$. We evaluate the likelihood using log-beta and minimize its negative using L-BFGS-B. The starting mean is the pooled adjustment rate $\sum_i A_i / \sum_i N_i$, and starting concentrations are 1, 5, and 20. We retain the solution with the highest finite log-likelihood, checking convergence and whether a numerical bound is reached. We used 705 pre-intervention carts from 478 non-member consumers in CFS-enabled regions. All three starts converge to the same optimum, with the marginal log-likelihood of -227.84. The estimated Beta distribution has parameters of $\hat{a} = 0.5455$ and $\hat{b} = 3.8652$.

EC.4.2. Gamma Distribution for Monthly Order-arrival Rates

We follow the notation in Section 5.2. Specifically, N_i is the number of completed pre-intervention orders and $T_i > 0$ is the observable tenure in months. We specify

$$N_i \mid \lambda_i, T_i \sim \text{Poisson}(\lambda_i T_i), \qquad \lambda_i \sim \text{Gamma}(r, \alpha), \qquad r, \alpha > 0.$$

We use the shape–rate parameterization: r is the shape and α is the rate. Its density is

$$f(\lambda \mid r, \alpha) = \frac{\alpha^r}{\Gamma(r)} \lambda^{r-1} e^{-\alpha\lambda}, \qquad \lambda > 0.$$

Consequently, $\mathbb{E}[\lambda_i] = r/\alpha$ and $\mathrm{Var}(\lambda_i) = r/\alpha^2$.

Integrating out the latent rate produces a negative-binomial probability for each observed order count conditional on its exposure:

$$\begin{aligned}\Pr(N_i \mid T_i, r, \alpha) &= \int_0^\infty \frac{e^{-\lambda T_i}(\lambda T_i)^{N_i}}{N_i!} f(\lambda \mid r, \alpha)\, d\lambda \\ &= \frac{\Gamma(N_i + r)}{\Gamma(r)\Gamma(N_i + 1)} \left(\frac{\alpha}{\alpha + T_i}\right)^r \left(\frac{T_i}{\alpha + T_i}\right)^{N_i}.\end{aligned}$$

This distribution has conditional mean $T_i r/\alpha$ and variance $T_i r/\alpha + T_i^2 r/\alpha^2$. The exposure term therefore accommodates different observable pre-intervention tenures.

Assuming independence across consumers, we maximize

$$\begin{aligned}\ell_G(r, \alpha) = \sum_{i \in C_G} \Big[&\log\Gamma(N_i + r) - \log\Gamma(r) - \log\Gamma(N_i + 1) \\ &+ r \log\alpha + N_i \log T_i - (N_i + r)\log(\alpha + T_i)\Big],\end{aligned} \tag{EC.4}$$

giving

$$(\hat{r}, \hat{\alpha}) \in \arg\max_{r>0,\, \alpha>0} \ell_G(r, \alpha).$$

We optimize over $(\log r, \log\alpha)$ using L-BFGS-B, evaluating log-gamma functions directly for numerical stability. We initialize r at 0.01, 0.1, 1, 10, and 100, and initialize $\alpha = r/\bar{\lambda}_{\text{pool}}$, where $\bar{\lambda}_{\text{pool}} = \sum_i N_i / \sum_i T_i$. The solution with the highest finite log-likelihood among converged starts is retained and checked for a numerical boundary solution. We use all consumers (even including those who browsed but not purchased) during pre-intervention periods. The final marginal log-likelihood is -2216.424, and the estimated Gamma distribution parameters are $\hat{r} = 10.583$ and $\hat{\alpha} = 86.176$.

EC.5. Budget-Constraints as a Complementary Mechanism.

In this section, we examine budget constraints as a complementary explanation for consumers' adverse responses to higher free-shipping requirements. This explanation concerns whether the additional spending is financially difficult for a consumer and can coexist with reference-dependent perceived attainability. Raising the threshold from \$80 to \$100 increases the spending required for free shipping by \$20. This incremental expenditure may be relatively easy to absorb for consumers with greater budget slack, but may carry a substantial opportunity cost, or simply be infeasible, for consumers with more limited purchasing power. Consequently, budget-constrained consumers may become less willing or able to adjust their carts and may instead place a smaller order, pay the shipping fee, postpone the purchase, or leave the platform.

To provide evidence, we use an indicator *HighAffluence* as a moderator. $HighAffluence_i = 1$ if the median after-tax household income of the consumer's delivery-address census tract exceeds the sample median, and zero otherwise. We then use the following triple DiD specification and report the regression results in Table EC.4.

$$y_{it} = \beta_0 + \beta_1 \cdot Treated_i \times Post_t + \beta_2 \cdot HighAffluence_i \times Treated_i \times Post_t + \beta_3 \cdot HighAffluence_i \times Post_t + \mu_i + \delta_t + \varepsilon_{it}$$

Table EC.4 Heterogeneous Treatment Effect by Consumer Affluence.

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.328**	-0.316**	-0.051**
	(0.131)	(0.128)	(0.021)
$Treated \times Post \times HighAffluence$	0.423**	0.400**	0.071**
	(0.198)	(0.192)	(0.030)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.031	0.030	0.031

*Notes: Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

For consumers residing in lower-affluence census tracts, the estimated effects of the threshold increase on spending, average cart value, and purchase frequency are -0.328, -0.316, and -0.051, respectively, and all three are statistically significant at the 5% level. More importantly, the corresponding $Treated \times Post \times HighAffluence$ coefficients are positive and statistically significant at the 5% level. Thus, the adverse response to the threshold increase is significantly attenuated among consumers living in higher-affluence areas. This pattern is consistent with the financial-feasibility account: consumers with greater local purchasing power appear better able to absorb the additional expenditure required to reach the new threshold.

We highlight, however, that the measures do not directly identify the mechanism at the individual level. Specifically, *HighAffluence* is based on median after-tax household income in the consumer's census tract. It therefore captures the purchasing power of the surrounding neighborhood rather than the consumer's actual income, liquidity, weekly grocery budget, or financial obligations, which may better proxy for the budget constraints at the individual level.

EC.5.1. Online Randomized Controlled Experiment

To complement our field evidence and examine whether budget constraints moderate responses to free-shipping requirements, we conducted a randomized online experiment on Prolific. Participants were randomly assigned to a 2 (*low gap* versus *high gap*: \$8 versus \$28) × 2 (*ample budget*

versus *tight budget*: \$180 versus \$120) between-subjects design. After excluding participants who failed attention checks or did not complete the experiment, the final analyzed sample comprised $N = 158$ participants: 40 in the *low-gap/ample-budget* condition, 39 in *low-gap/tight-budget*, 40 in *high-gap/ample-budget*, and 39 in *high-gap/tight-budget*.

Participants were asked to imagine shopping at FreshMarket Online, a hypothetical online grocery platform. All monetary values used in the scenario were pretested. Participants were told that they had either \$180 or \$120 available for all grocery purchases for the rest of the week and expected another grocery purchase of approximately \$30 later that week. A free-shipping threshold of \$80 or \$100 required adding \$8 or \$28 to the current cart, respectively. The initial cart value (\$72) and the below-threshold shipping fee (\$5.99) were held constant across all four conditions. Participants chose among three options: (1) continue shopping and check out with free shipping; (2) pay the \$5.99 shipping fee and check out with the current cart; or (3) leave the cart without checking out at that time.

After making their checkout choice, participants completed a questionnaire assessing their perceptions of the shopping situation and free-shipping requirement and provided demographic information. We measured opportunity-cost consideration to examine whether a larger gap to the CFS threshold under tight budgets draws attention to other grocery purchases consumers might have to forgo. Following Spiller (2011), we measured opportunity-cost consideration using three items adapted to the focal grocery-shopping decision: (1) "While deciding what to do, I thought about how spending the extra amount needed to qualify for free shipping would leave less money for other purchases later"; (2) "While deciding, I thought about other groceries I could buy later this week with the money I would need to spend to qualify for free shipping"; and (3) "I considered specific purchases I might have to postpone, reduce, or give up if I spent the extra amount needed to qualify for free shipping." Responses ranged from 1 (strongly disagree) to 6 (strongly agree). The three items were averaged to form an opportunity-cost consideration score (Cronbach's $\alpha = 0.77$), with higher scores indicating greater opportunity-cost consideration.

We assessed the manipulations using two questions. The first assessed whether the budget manipulation changed perceived budget tightness: "The total amount available for grocery purchases this week felt tight" (1 means strongly disagree; 7 means strongly agree). The second assessed whether the gap manipulation changed the perceived size of the additional spending requirement: "How large did the additional amount needed to receive free shipping seem?" (1 means very small; 7 means very large).

Overall, 47.5% of participants reported being female, 61.4% identified as White, and 59.5% reported full- or part-time employment. Mean age was 40.1 years ($SD = 12.9$). Balance checks did not detect statistically significant differences in the measured demographic characteristics across the four conditions.

Reported means and percentages below are unadjusted descriptive statistics. We estimated a logistic regression for the choice to continue shopping and check out with free shipping containing the main effects of gap and available grocery budget and their interaction, controlling for age categories, gender, White ethnicity, and employment status. Opportunity-cost consideration was analyzed using ordinary least squares regressions containing the two experimental main effects and their interaction. Our main results are as follows.

Participants assigned to a *tight budget* reported greater perceived budget tightness than those assigned to an *ample budget* (4.99 versus 3.73; $p < 0.001$). Participants assigned to the *high gap* rated the additional spending requirement as larger than those assigned to the *low gap* (4.54 versus 2.95; $p < 0.001$). These results indicate that both manipulations changed perceptions in the intended direction.

We first examined the distribution of the three checkout choices. In the *low-gap/ample-budget* condition, 82.5% chose to continue shopping and check out with free shipping, 17.5% selected checkout with paid shipping, and none selected leaving without placing the order. The corresponding percentages were 94.9%, 5.1%, and 0.0% in *low-gap/tight-budget*; 62.5%, 27.5%, and 10.0% in *high-gap/ample-budget*; and 46.2%, 43.6%, and 10.3% in *high-gap/tight-budget*. A Pearson chi-square test indicated that the overall choice distribution differed across the four conditions ($\chi^2(6) = 28.845$, $p < 0.001$).

We then estimated a logistic regression, coding the choice to continue shopping and check out with free shipping as one and either alternative as zero. *High gap* and *tight budget* were each coded as one, with *low gap* and *ample budget* coded as zero. The *high gap* × *tight budget* interaction was negative ($b = -2.633$, $SE = 1.016$, $p = 0.010$), indicating that the larger gap's effect on the log odds of selecting this option was more negative under a tight budget.

Mean opportunity-cost consideration was 3.81, 3.37, 3.94, and 4.68 in the *low-gap/ample-budget*, *low-gap/tight-budget*, *high-gap/ample-budget*, and *high-gap/tight-budget* conditions, respectively. The *high gap* × *tight budget* interaction was positive ($b = 1.183$, $p = 0.001$), indicating that the increase in opportunity-cost consideration associated with the larger gap was greater under a tight

budget. The interaction remained positive and statistically significant after adding demographic controls ($b = 1.278$, $p < 0.001$).

The experimental findings suggest that pursuing free shipping can compete with other planned grocery expenditures. Together, the field analysis and randomized experiment provide evidence consistent with budget constraints as a complementary explanation for adverse responses to higher free-shipping requirements.

EC.6. Counterfactual Analysis and Threshold Design

EC.6.1. Proof of Proposition 1

Fix $c < 0$, $C > 0$, $k \in [0, 1)$, and $\underline{\tau} < \overline{\tau}$. Let the average cart value under threshold τ be

$$y(\tau) = a\left(\frac{\tau}{\underline{\tau}}\right)^c, \qquad \tau \in [\underline{\tau}, \overline{\tau}],$$

where $a = y(\underline{\tau})$ is given. The platform's per-order net revenue is

$$\pi(\tau) = y(\tau) + S(y(\tau), \tau) - C,$$

where

$$S(y, \tau) = \begin{cases} C - ky, & y < \tau, \\ 0, & y \geq \tau. \end{cases}$$

We first establish the basic monotonicity of demand. Differentiating $y(\tau)$ yields

$$y'(\tau) = \frac{c}{\tau} y(\tau),$$

which is strictly negative for all $\tau > 0$ because $c < 0$. Hence $y(\tau)$ is strictly decreasing in τ on $[\underline{\tau}, \overline{\tau}]$.

Next, consider the equation $y(\tau) = \tau$. Substituting the expression for $y(\tau)$ gives

$$a\left(\frac{\tau}{\underline{\tau}}\right)^c = \tau \quad \Longleftrightarrow \quad \tau^{1-c} = a\,\underline{\tau}^{-c}.$$

Since $1 - c > 1$, this equation admits a unique positive solution,

$$\hat{\tau}(a) = \left(a\,\underline{\tau}^{-c}\right)^{\frac{1}{1-c}}.$$

Because $y(\tau)$ is strictly decreasing while τ is increasing, we have $y(\tau) > \tau$ if and only if $\tau < \hat{\tau}(a)$ and $y(\tau) < \tau$ if and only if $\tau > \hat{\tau}(a)$.

Using this characterization, the profit function can be written as

$$\pi(\tau) = \begin{cases} y(\tau) - C, & \tau \leq \hat{\tau}(a), \\ (1 - k)y(\tau), & \tau > \hat{\tau}(a). \end{cases}$$

On the interval $\tau \le \hat{\tau}(a)$, i.e., the free shipping region, $\pi(\tau) = y(\tau) - C$ is strictly decreasing in τ. On the interval $\tau > \hat{\tau}(a)$, i.e., the surcharge region, $\pi(\tau) = (1-k)y(\tau)$ is also strictly decreasing in τ because $1-k > 0$ and $y(\tau)$ is strictly decreasing. Therefore, within each region the profit function is strictly decreasing in τ.

It follows that any optimal threshold must be attained either at the lower bound $\underline{\tau}$ or at the boundary between the two regions, provided that boundary is feasible. If $\hat{\tau}(a) > \overline{\tau}$, then $y(\tau) \ge \tau$ for all $\tau \in [\underline{\tau}, \overline{\tau}]$ and the entire feasible set lies in the free-shipping region. Since $\pi(\tau) = y(\tau) - C$ is strictly decreasing on this interval, the unique maximizer is $\tau^* = \underline{\tau}$.

Suppose instead that $\hat{\tau}(a) \le \overline{\tau}$. In this case, both the free-shipping region and the surcharge region are feasible. The best feasible threshold in the free-shipping region is $\underline{\tau}$, yielding profit

$$\pi(\underline{\tau}) = y(\underline{\tau}) - C = a - C.$$

In the surcharge region, because $\pi(\tau)$ is strictly decreasing for $\tau > \hat{\tau}(a)$, the largest attainable profit is achieved in the limit as τ approaches $\hat{\tau}(a)$ from above. By continuity of $y(\tau)$,

$$\lim_{\tau \downarrow \hat{\tau}(a)} \pi(\tau) = (1-k)y(\hat{\tau}(a)) = (1-k)\hat{\tau}(a).$$

Thus, when $\hat{\tau}(a) \le \overline{\tau}$, the platform prefers the kink to the lower bound if and only if

$$(1-k)\hat{\tau}(a) \ge a - C.$$

Define $a^* > 0$ as the unique solution to

$$(1-k)\hat{\tau}(a^*) = a^* - C.$$

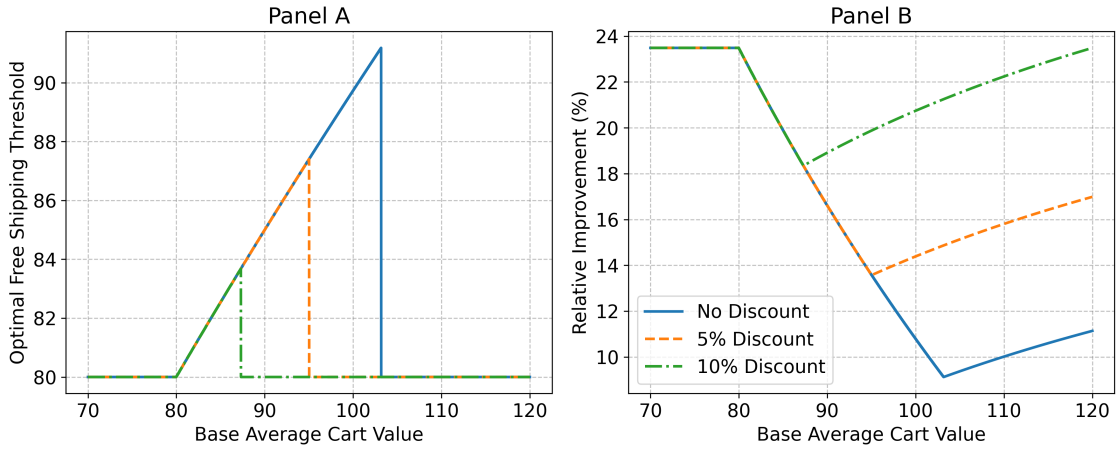
Such a solution exists because $\hat{\tau}(a)$ is continuous and strictly increasing in a, grows sublinearly in a, and satisfies $\hat{\tau}(\underline{\tau}) = \underline{\tau}$.

Therefore, if $a \in [\underline{\tau}, a^*]$ and $\hat{\tau}(a) \le \overline{\tau}$, the optimal threshold is $\tau^* = \hat{\tau}(a)$ (interpreted as the right-limit at the kink). In all other cases, the optimal threshold is $\tau^* = \underline{\tau}$. This completes the proof.

EC.6.2. Impact of Shipping Discount

We also vary the discount factor k and investigate the performance of a partially discounted CFS policy. We set $k = 0$ (no discounting), $k = 0.05$ (small discounting), and $k = 0.1$ (large discounting). The results are shown in Figure EC.2. We find that as the discounting factor increases, the support for an optimal threshold above 80 shrinks (Panel A). In addition, the minimum relative improvement increases, indicating that offering consumers a larger discount on the shipping surcharge makes suboptimal threshold choices more costly.

Figure EC.2 Optimal Threshold Values and Revenue Improvement with Various Base Cart Values and Shipping Discounts



Panel A plots the optimal free-shipping threshold as a function of the base average cart value. Panel B reports the relative improvement in net consumer revenue (%) when the threshold is set to its optimal value compared with a fixed threshold of $100, for each base average cart value.

EC.6.3. Stochastic Optimization with Cart Value Heterogeneity

This section extends the deterministic threshold optimization problem by allowing the original cart values and the elasticity to be heterogeneous across consumers and orders. Let $a \equiv y(\underline{\tau})$ denote the cart value at the original free shipping threshold $\underline{\tau}$, and let c denote the (order-level) elasticity of cart value with respect to the threshold. We assume that (a, c) is a random vector on $\mathbb{R}_+ \times \mathbb{R}$ that jointly follows distribution $\mathcal{F}$, with the cumulative distribution function $F(\cdot, \cdot)$.

For any candidate threshold $\tau \in [\underline{\tau}, \overline{\tau}]$, realized cart value is given by

$$y(\tau; a, c) = a\left(\frac{\tau}{\underline{\tau}}\right)^c,$$

An order qualifies for free shipping if and only if $y(\tau; a, c) \geq \tau$, which is equivalent to

$$a \geq a_0(\tau; c) \equiv \tau^{1-c}\underline{\tau}^c.$$

The platform's per-cart net revenue from an order with baseline cart value a is

$$\pi(\tau; a, c) = \begin{cases} y(\tau; a, c) - C, & a \geq a_0(\tau; c), \\ (1-k)y(\tau; a, c), & a < a_0(\tau; c), \end{cases}$$

where C is the per-cart fulfillment cost and $k \in [0, 1)$ governs how the shipping surcharge varies with cart value. The platform chooses τ to maximize expected per-cart profit,

$$\Pi(\tau) \equiv \mathbb{E}_{\mathcal{F}}[\pi(\tau; a, c)], \quad \tau^* \in \arg\max_{\underline{\tau} \leq \tau \leq \overline{\tau}} \Pi(\tau)$$

ASSUMPTION EC.1. *(1) The elasticity c is a constant and negative. (2) The original cart value follows a uniform distribution, $a \sim Unif[L, U]$, where $0 < L < U$. (3) $L < \underline{\tau} < U$.*

Note that the first assumption is supported by the estimation results from a random coefficient model in Section EC.6.3.1. The second assumption simplifies the analysis. Despite its simplicity, the uniform case highlights that the optimal free-shipping threshold depends not only on average cart value but also on its dispersion.

Let $g(\tau) \equiv (\tau/\underline{\tau})^c$. Under the above assumption, we have

$$y(\tau; a, c) = g(\tau)a, \qquad F(t) = \frac{t-L}{U-L}, \qquad \mathbb{E}[a\,\mathbf{1}\{a \geq t\}] = \frac{U^2 - t^2}{2(U-L)}.$$

Substituting $t = a_0(\tau)$ yields

$$\Pi(\tau) = g(\tau)\left[(1-k)\frac{L+U}{2} + k\frac{U^2 - a_0(\tau)^2}{2(U-L)}\right] - C\left(1 - \frac{a_0(\tau) - L}{U-L}\right),$$

for $a_0(\tau) \in (L, U)$. Let $\mu \equiv \mathbb{E}[a] = (L+U)/2$. When $a_0(\tau) \leq L$, all orders qualify for free shipping and $\Pi(\tau) = g(\tau)\mu - C$; when $a_0(\tau) \geq U$, no orders qualify and $\Pi(\tau) = (1-k)g(\tau)\mu$.

Because $g(\tau)$ is strictly decreasing in τ, expected profit is decreasing in τ outside the interior region $a_0(\tau) \in (L, U)$. Hence, an interior optimum (if it exists) must lie in the mixed regime where some orders qualify for free shipping and others incur a surcharge. In this region, the optimal threshold trades off the marginal loss from reduced demand against the marginal gain from shifting probability mass from free shipping to surcharge, with the density of orders around $a_0(\tau)$ playing a central role.

PROPOSITION EC.1 **(Stochastic Optimal Free-Shipping Thresholds)**. *Let $D \equiv U - L$ and $\mu \equiv \mathbb{E}[a] = (L+U)/2$ under Assumption EC.1. Define the cutoff*

$$\mu^* \equiv \frac{1}{1-k}\left[\frac{1-c}{-c} \cdot \frac{\underline{\tau}}{D}\left(C - k\underline{\tau}\right) - k \cdot \frac{U^2 - \underline{\tau}^2}{2D}\right].$$

If $\mu < \mu^*$, then the platform optimally increases *the threshold above $\underline{\tau}$.*

The proof can be seen in EC.6.3.2. We highlight that Proposition EC.1 naturally extends our findings in the main body (Proposition 1) to a stochastic setting with mild assumptions. In the stochastic model, the platform chooses a single threshold τ for a population of heterogeneous orders. Increasing τ raises $a_0(\tau)$, thereby shifting some orders from the free shipping regime into the surcharge regime. Unlike the deterministic case, the marginal benefits depends on the amount

of probability mass near the cutoff $a_0(\tau)$, i.e., on the distribution of original cart value $\mathcal{F}$, not just its mean. Under Assumption EC.1, the strength of this shift is reciprocal to dispersion D. This is why the cutoff μ^* in Proposition EC.1 depends on both the mean μ and dispersion D. In particular, holding μ fixed, greater dispersion lowers the density around the cutoff and weakens the incentive to raise the threshold τ.

Figure EC.3 Optimal Threshold Values with Various Base Cart Values

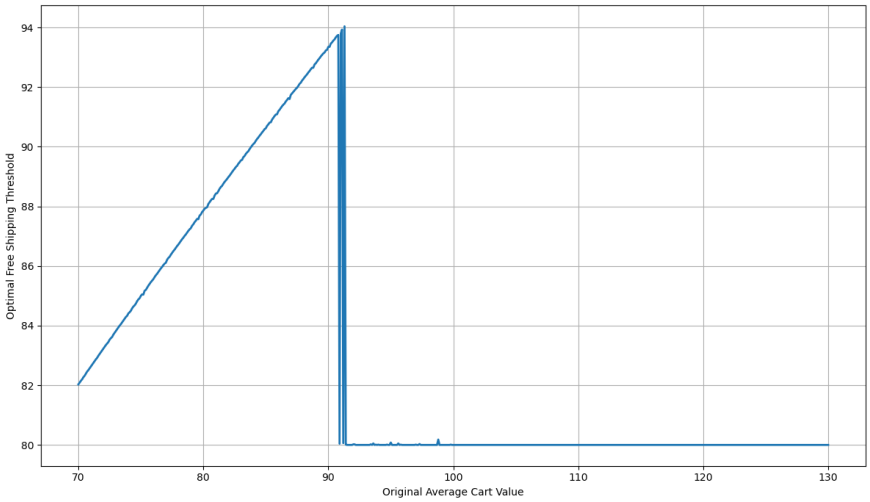


Figure EC.3 illustrates these results numerically. We assume that pre-policy cart values follow a uniform distribution, $a \sim \text{Unif}[0.8\mu, 1.2\mu]$, where $\mu = \mathbb{E}[a]$ varies from 70 to 130. For each value of μ, we draw 10,000 realizations of a and compute the optimal free-shipping threshold by maximizing empirical per-cart profit. Consistent with Proposition EC.1, the optimal policy exhibits a cutoff structure: increasing the threshold above the baseline is optimal only for μ below a critical value, while for larger μ the optimal threshold remains at the lower bound and should not be increased further.

EC.6.3.1. A random-slope DiD model. We assume the treatment effect is random at the consumer level, and estimate the coefficients with the following DiD specification:

$$\log(AvgCartValue + 1) = \alpha_0 + (\alpha_1 + \alpha_{1,i}) \cdot Treated_i \times Post_t + \mu_i + \delta_t + \varepsilon_{it}$$

where α_1 captures the common treatment effect, and $\alpha_{1,i} \sim \mathcal{N}(0, \sigma^2)$ captures the random heterogeneous treatment effects across consumers. Table EC.5 reports the results. In particular, $\alpha_1 = -0.211$ and $\sigma^2 \simeq 0$, which support the the first part in Assumption EC.1.

Table EC.5 Random Effects of CFS Threshold Increase on Average Cart Value

	(1) *AvgCartValue*
$Treated \times Post$	-0.211**
	0.098
σ^2	1.63e-11
Consumer Fixed Effects	Yes
Yearmonth Fixed Effects	Yes
Observations	5148
Log-likelihood	-10464.43

*Notes: Standard errors are clustered at the consumer-level. * p<0.10, ** p<0.05, *** p<0.01.*

EC.6.3.2. Proof of Proposition EC.1. Fix $\tau \in [\underline{\tau}, \overline{\tau}]$. Note that $g(\tau) \equiv \left(\frac{\tau}{\underline{\tau}}\right)^c$ and $a_0(\tau) \equiv \tau^{1-c}\underline{\tau}^c$; hence, $y(\tau;a) = g(\tau)a$ and $y(\tau;a) \geq \tau$ if and only if $a \geq a_0(\tau)$. Note that $g(\tau) > 0$ and, since $c < 0$, $g'(\tau) = \frac{c}{\tau}g(\tau) < 0$, while $a_0'(\tau) = \frac{1-c}{\tau}a_0(\tau) > 0$. Moreover,

$$g(\tau)\, a_0(\tau) = \tau. \tag{EC.5}$$

The platform maximizes expected per-order profit $\Pi(\tau) \equiv \mathbb{E}[\pi(\tau;a)]$, where

$$\pi(\tau;a) = \begin{cases} y(\tau;a) - C, & a \geq a_0(\tau), \\ (1-k)y(\tau;a), & a < a_0(\tau). \end{cases}$$

Using $y(\tau;a) = g(\tau)a$ and $a \sim \text{Unif}[L,U]$, we obtain

$$\begin{aligned} \Pi(\tau) &= \mathbb{E}\big[(g(\tau)a - C)\mathbf{1}\{a \geq a_0(\tau)\} + (1-k)g(\tau)a\mathbf{1}\{a < a_0(\tau)\}\big] \\ &= g(\tau)\Big((1-k)\mathbb{E}[a] + k\,\mathbb{E}[a\mathbf{1}\{a \geq a_0(\tau)\}]\Big) - C\,\mathbb{P}(a \geq a_0(\tau)). \end{aligned}$$

In the mixed region $a_0(\tau) \in (L,U)$, the uniform distribution yields

$$\mathbb{P}(a \geq t) = \frac{U-t}{D}, \qquad \mathbb{E}[a\mathbf{1}\{a \geq t\}] = \frac{U^2 - t^2}{2D}.$$

Substituting $t = a_0(\tau)$ gives, for $a_0(\tau) \in (L,U)$,

$$\Pi(\tau) = g(\tau)\left[(1-k)\mu + k\frac{U^2 - a_0(\tau)^2}{2D}\right] - C\frac{U - a_0(\tau)}{D}. \tag{EC.6}$$

We next compute the derivative at the baseline threshold $\underline{\tau}$. Since $\underline{\tau}$ lies in the mixed region, (EC.6) is differentiable at $\underline{\tau}$. Differentiating (EC.6) and using $g'(\tau) = \frac{c}{\tau}g(\tau)$, $a_0'(\tau) = \frac{1-c}{\tau}a_0(\tau)$, and the identity (EC.5), yields

$$\Pi'(\tau) = \frac{1}{\tau}\left\{c\, g(\tau)\left[(1-k)\mu + k\frac{U^2 - a_0(\tau)^2}{2D}\right] + (1-c)\frac{a_0(\tau)}{D}\big(C - k\tau\big)\right\}, \qquad a_0(\tau) \in (L,U).$$

Evaluating at $\tau = \underline{\tau}$ gives $g(\underline{\tau}) = 1$ and $t(\underline{\tau}) = \underline{\tau}$, hence

$$\Pi'(\underline{\tau}) = \frac{1}{\underline{\tau}} \left\{ c \left[(1-k)\mu + k\frac{U^2 - \underline{\tau}^2}{2D} \right] + (1-c)\frac{\underline{\tau}}{D}\left(C - k\underline{\tau}\right) \right\}. \tag{EC.7}$$

Because $\underline{\tau} > 0$, the sign of $\Pi'(\underline{\tau})$ equals the sign of the bracketed expression. Rearranging (EC.7) shows that $\Pi'(\underline{\tau}) \le 0$ if and only if $\mu \ge \mu^*$, where

$$\mu^* = \frac{1}{1-k}\left[\frac{1-c}{-c} \cdot \frac{\underline{\tau}}{D}\left(C - k\underline{\tau}\right) - k \cdot \frac{U^2 - \underline{\tau}^2}{2D} \right].$$

Finally, since $\Pi(\tau)$ is continuous on the compact set $[\underline{\tau}, \overline{\tau}]$, a global maximizer τ^* exists. If $\Pi'(\underline{\tau}) > 0$, then $\Pi(\tau) > \Pi(\underline{\tau})$ for τ in a neighborhood above $\underline{\tau}$, so $\underline{\tau}$ cannot be the global maximizer and $\tau^* > \underline{\tau}$. Combining with the equivalence $\Pi'(\underline{\tau}) > 0 \Longleftrightarrow \mu < \mu^*$ yields

$$\mu < \mu^* \implies \tau^* > \underline{\tau}.$$

This completes the proof.

EC.6.4. DiD Analysis with Continuous Interpurchase Time

We denote InterTime_i as a continuous variable, indicating the time difference between consumer i's last pre-treatment purchase and the threshold change. We run the following DiD specification. The estimated results are shown in Table EC.6.

$$\begin{aligned}\log(\text{AvgCartValue}_{it} + 1) = {} & \beta_0 + \mu_i + \delta_t + \beta_1(\text{Treated} \times \text{Post}) \\ & + \beta_2 \text{Treated}_i \times \text{Post}_t \times \log(\text{InterTime}_i) + \beta_3 \text{Post}_t \times \log(\text{InterTime}_i)\end{aligned}$$

Table EC.6 Effects of CFS Threshold Increase by InterTime.

	AvgCartValue
Treated × Post	-0.674***
	(0.158)
Treated × Post × log(InterTime)	0.380***
	(0.080)
Consumer Fixed Effects	Yes
Yearmonth Fixed Effects	Yes
Observations	5148
Adjusted R^2	0.03

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01.*

EC.7. Estimation Results for Robustness Checks

EC.7.1. Parallel Trends Assumption

Table EC.7 Regression Results for the Parallel Trends Assumption Tests.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times 202302$	0.359	0.354	0.060	0.160	0.143	0.047
	(0.417)	(0.415)	(0.063)	(0.160)	(0.260)	(0.063)
$Treated \times 202303$	0.056	0.045	0.018	0.090	-0.079	-0.013
	(0.426)	(0.423)	(0.062)	(0.164)	(0.282)	(0.062)
$Treated \times 202304$	-0.293	-0.295	-0.041	-0.066	-0.296	-0.079
	(0.411)	(0.406)	(0.062)	(0.158)	(0.270)	(0.060)
$Treated \times 202305$	0.478	0.471	0.077	0.216	0.220	0.037
	(0.412)	(0.407)	(0.061)	(0.173)	(0.241)	(0.054)
$Treated \times 202306$	0.188	0.183	0.034	0.087	0.049	0.032
	(0.396)	(0.392)	(0.060)	(0.161)	(0.241)	(0.059)
$Treated \times 202307$	0.100	0.106	0.014	0.122	-0.014	-0.019
	(0.373)	(0.372)	(0.055)	(0.149)	(0.227)	(0.053)
$Treated \times 202308$	-0.054	-0.041	-0.005	-0.009	-0.036	-0.030
	(0.388)	(0.387)	(0.056)	(0.162)	(0.232)	(0.053)
$Treated \times 202309$	-0.238	-0.239	-0.037	-0.057	-0.221	-0.057
	(0.420)	(0.418)	(0.061)	(0.170)	(0.259)	(0.060)
$Treated \times 202310$	0.242	0.227	0.044	0.035	0.177	0.006
	(0.446)	(0.442)	(0.068)	(0.175)	(0.289)	(0.066)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Consumers	261	261	261	261	261	261
Observations	2277	2277	2277	2277	2277	2277
Adjusted R^2	0.034	0.032	0.040	0.078	0.023	0.042

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except PopShare, are log-transformed. We use only the pre-intervention observations.*

EC.7.2. Placebo Tests

Table EC.8 Regression Results for the Placebo Tests.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
Panel A: Placebo test with fake CFS threshold increase.						
$Treated \times PlaceboPost$	-0.074	-0.069	-0.013	-0.046	-0.006	-0.011
	(0.204)	(0.201)	(0.031)	(0.082)	(0.125)	(0.030)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Consumers	261	261	261	261	261	261
Observations	2277	2277	2277	2277	2277	2277
Adjusted R^2	0.034	0.032	0.040	0.079	0.023	0.041
Panel B: Placebo test with fake treatment group						
$PlaceboTreated \times Post$	-0.026	-0.030	-0.007	0.023	-0.038	-0.013
	(0.152)	(0.147)	(0.024)	(0.058)	(0.097)	(0.022)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Consumers	129	129	129	129	129	129
Observations	2549	2549	2549	2549	2549	2549
Adjusted R^2	0.031	0.031	0.028	0.053	0.025	0.038

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except PopShare, are log-transformed. Panel A uses only the pre-intervention observations; Panel B uses only the observations of the control group.*

EC.7.3. Alternative Measures for Reference Point Formation

To examine whether the heterogeneity results depend on how we measure prior purchasing experience, we re-estimate the same specification using two alternative moderators. First, we use the average pre-intervention transactions per month, calculated as N_i/T_i, as the measure. We take its logarithm and standardize it to construct the moderator:

$$PreTransRate_i = \frac{\log(N_i/T_i) - \overline{\log(N_i/T_i)}}{s_{\log(N_i/T_i)}}$$

Second, we use the standardized logarithm of the total number of completed pre-intervention orders

$$Exposure_i = \frac{\log(N_i) - \overline{\log(N_i)}}{s_{\log(N_i)}}$$

which captures the cumulative shopping experience regardless of timing. We treat each alternative measure as a continuous moderator and replace $Formation$ with the alternatives in the above specification. Tables EC.9 and EC.10 show consistent conclusions.

Table EC.9 Heterogeneous Effects by Pre-intervention Monthly Transaction Rate

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	0.221**	0.214**	0.037***
	(0.090)	(0.088)	(0.014)
$Treated \times Post \times PreTransRate$	-1.123***	-1.089***	-0.178***
	(0.130)	(0.128)	(0.020)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.042	0.041	0.041

*Notes: Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

Table EC.10 Heterogeneous Effects by Cumulative Pre-intervention Purchasing Exposure

	Spending	*AvgCartValue*	*Transactions*
$Treated \times Post$	-0.175*	-0.171*	-0.025*
	(0.092)	(0.090)	(0.014)
$Treated \times Post \times Exposure$	-0.420***	-0.402***	-0.070***
	(0.060)	(0.060)	(0.009)
Consumer Fixed Effects	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes
Observations	5148	5148	5148
Adjusted R^2	0.039	0.038	0.038

*Notes: Standard errors are clustered at the consumer level. All outcome variables are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*

EC.7.4. Coarsened Exact Matching (CEM)

Because consumers are not randomly assigned to the treatment and control groups, their shopping behaviors may differ systematically, potentially violating the parallel trends assumption required for the DiD identification. To address that, we first show that consumers' browsing and purchasing behaviors are largely statistically indistinguishable between the treatment and control groups prior to the intervention. We consider six outcome variables described in Section 3.2.3, along with four additional measures of browsing behavior. Specifically, *Session* denotes the number of sessions a consumer has in each month; *AvgViews* is the average number of distinct products viewed per session; *AvgSessionDuration* measures the average session length in seconds; and *Conversion* is the proportion of viewed products that result in a purchase. As reported in Table EC.11, none of these variables exhibit statistically significant differences across groups, with the exception of average cart value, which is marginally different.

In addition, we implement coarsened exact matching (CEM) (Iacus et al. 2009), a nonparametric matching method that allows researchers to pre-specify the level of acceptable imbalance between treated and control units and has been commonly used in DiD applications (e.g., He et al. 2023, Kumar et al. 2019, Mejia et al. 2021). We use *Spending*, *AvgCartValue*, *Variety*, and *Conversion* as matching covariates. These variables are selected based on their theoretical relevance for post-treatment purchasing behavior and their relatively low redundancy in the data's correlation structure. Together, they capture the primary dimensions of consumers' purchasing intensity, basket composition, and purchase propensity, while avoiding excessive overlap with highly collinear outcome variables and browsing measures.

We coarsen each covariate into a small number of substantively meaningful bins to maximize common support. Specifically, *Spending* is grouped into three categories (low, medium, and high spenders); *AvgCartValue* into two categories (low versus high); *Variety* into two categories (low

Table EC.11 Covariate Balance Between Treatment and Control Groups Before Matching.

	Treatment Group		Control Group			
Variable	Mean	Std	Mean	Std	t-stats	p-value
Spending	5.306	0.773	5.373	0.794	-0.689	0.491
AvgCartValue	4.760	0.474	4.882	0.536	-1.942	0.053*
Transactions	1.044	0.396	1.002	0.343	0.922	0.357
Variety	1.980	0.884	2.069	0.911	-0.801	0.424
AvgPrice	3.465	0.826	3.465	0.692	-0.007	0.995
PopShare	0.504	0.176	0.496	0.181	0.348	0.728
Sessions	1.279	0.572	1.355	0.706	-0.959	0.338
AvgViews	0.897	0.715	0.933	0.740	-0.398	0.691
AvgSessionDuration	6.620	1.447	6.642	1.510	-0.122	0.903
Conversion	0.572	0.159	0.553	0.166	0.928	0.354

*Notes: * p<0.10, ** p<0.05, *** p<0.01.*

Table EC.12 The Quality for the CEM Matching

	SMD	
Covariate	**All Samples**	**Matched Samples**
Spending	-0.130	-0.064
AvgCartValue	0.019	0.031
Variety	-0.121	0.060
Conversion	0.019	-0.015

versus high); and *Conversion* into two categories (low versus high). This coarse partitioning preserves a large share of treated and control observations in the matched sample while ensuring balance along key pre-treatment behavioral dimensions.

The resulting matched sample exhibited strong overlap and excellent covariate balance. We retained 83% of the sample and dropped 19 consumers from the treatment group and 25 from the control group. We report the standardized mean differences (SMD) between groups on all matched covariates to demonstrate the quality of the matching (Table EC.12). The SMD values remain below 0.1, well within conventional thresholds for acceptable balance.

Table EC.13 reports treatment effect estimates from the weighted DiD analysis. The results are consistent with our main findings and provide additional support for the robustness of our conclusions.

Table EC.13 DiD Analysis on Matched Samples.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times Post$	-0.326***	-0.376**	-0.062**	-0.138**	-0.230**	-0.057**
	(0.092)	(0.149)	(0.024)	(0.061)	(0.093)	(0.023)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	4280	4280	4280	4280	4280	4280
Adjusted R^2	0.01	0.01	0.01	0.01	0.02	0.02

*Notes: Standard errors are clustered at the consumer level. * p<0.10, ** p<0.05, *** p<0.01. All outcome variables, except PopShare, are log-transformed. 19 consumers in the treatment group and 25 in the control group are dropped due to matching, resulting in a smaller sample size.*

EC.7.5. Inverse Probability Weighting (IPW) Estimator

A potential concern with our main analysis is the possibility of selection bias arising from restricting the sample to consumers who made at least one purchase both before and after the intervention. To mitigate this potential bias, we adopt an inverse probability weighting (IPW) estimator (Robins et al. 1994). The key idea is to reweight the observed sample to resemble the full population that includes both retained and non-returned consumers. In our context, IPW accounts for the possibility that some consumers, especially those negatively affected by the policy change, may have stopped purchasing altogether after the intervention.

Table EC.14 The Quality for the IPW weights

Covariate	SMD	
	Unweighted	Weighted
AvgCartValue	0.281	0.012
Transactions	0.215	0.004
AvgPrice	-0.075	-0.031
Variety	0.317	0.006
PopShare	0.096	0.001
Sessions	0.138	-0.002
AvgViews	0.206	0.025
AvgSessionDuration	0.253	0.020
Conversion	-0.187	0.005

Specifically, we first estimate the probability that a consumer remains active after the intervention based on their pre-intervention aggregation-level characteristic vector Z_i. We use sample of consumers who placed at least one order before the intervention, which includes consumers who has not placed any order after the intervention ($Survive_i = 0$) and those who have placed at least one order ($Survive = 1$). We use a logit regression to estimate the probability that a consumer will place orders after the intervention based on pre-intervention purchasing and browsing behavior. Specifically,

$$\begin{aligned} logit\left(\mathbb{P}(Survive_i = 1 \mid Z_i)\right) = \alpha_0 + \alpha_1 \cdot AvgCartValue_i + \alpha_2 \cdot Transactions_i \\ + \alpha_3 \cdot AvgPrice_i + \alpha_4 \cdot Variety_i + \alpha_5 \cdot PopShare_i + \alpha_6 \cdot Sessions_i \\ + \alpha_7 \cdot AvgViews_i + \alpha_8 \cdot AvgSessionDuration_i + \alpha_9 \cdot Conversion_i \end{aligned} \quad \text{(EC.8)}$$

We then construct stabilized IPW weights w_i (Robins et al. 2000) for each consumer as:

$$w_i = \frac{\mathbb{P}(Survive_i = 1)}{\mathbb{P}(Survive_i = 1 \mid Z_i)} \quad \text{(EC.9)}$$

where $\mathbb{P}(Survive_i = 1)$ is the proportion of consumers who survived in the sample to estimate the logit regression. The numerator stabilizes the weights and reduces their variance by multiplying the estimated inverse probability by the overall retention probability.

We further report the standardized mean differences (SMD) between groups on all covariates before and after weighting to demonstrate the quality of the probabilities and the balance they deliver once turned into weights. As shown in Table EC.14, the absolute SMD before weighting is approximately 0.32, while it significantly reduced after weighting, implying the balance of the IPW weights.

Using these stabilized weights, we re-estimate our DiD model in Specification (1). Table EC.15 summarizes the estimation results.

Table EC.15 Regression Results Using Inverse Probability Weighting.

	(1) *Spending*	(2) *AvgCartValue*	(3) *Transactions*	(4) *Variety*	(5) *AvgPrice*	(6) *PopShare*
$Treated \times Post$	-0.274**	-0.265**	-0.041**	-0.101**	-0.137*	-0.034*
	(0.120)	(0.118)	(0.019)	(0.046)	(0.078)	(0.019)
Consumer Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Yearmonth Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5148	5148	5148	5148	5148	5148
Adjusted R^2	0.02	0.02	0.01	0.03	0.01	0.03

*Notes: Standard errors are clustered at the consumer level. All outcome variables, except PopShare, are log-transformed. * p<0.10, ** p<0.05, *** p<0.01.*